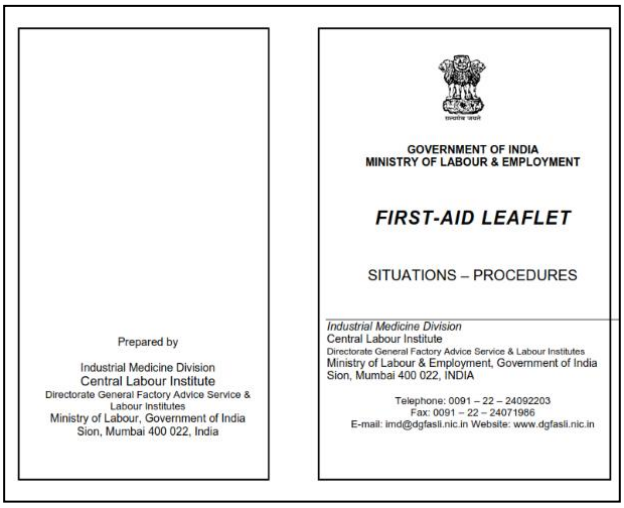
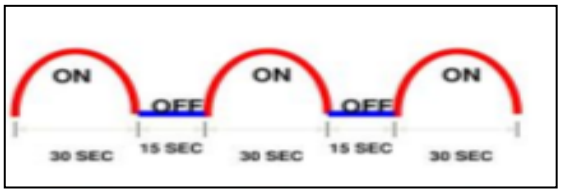
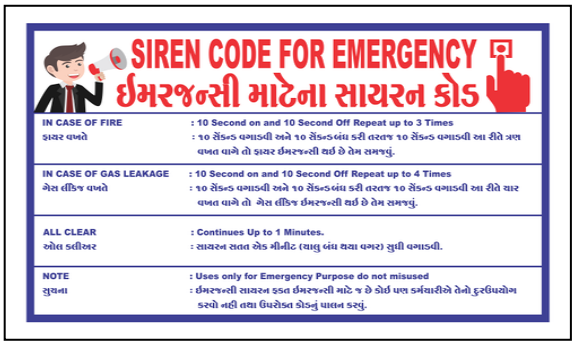
8. AUDIT OBSERVATION AND RECOMMANDATIONS

8.1 Annexure - A: Areas of Improvement

(Note- All Pics in recommendation are as reference and for better clarity only)

Sr. No.	Observation	Recommendation
Legal / General		
1.	"Abstract of the Factories Act, 1948 and The Gujarat factories Rules, 1963" (Form No. 23) was not displayed in a conspicuous and convenient place at or near the main entrance of the factory.	As per under Rule 106 of The Gujarat Factories Rules (GFR), 1963, "Abstract of the Factories Act, 1948 and The Gujarat factories Rules, 1963" (Form No. 23) should be displayed in a conspicuous and convenient place at or near the main entrance of the factory in prominent manner.
2.	First aid box available across the factory but copy of FIRST-AID LEAFLET (Situations-Procedures) issued by the Directorate general of Factories Advice Service and labour Institutes, Government of India, Mumbai was not evident in all First Aid Boxes.	As per the Rule 70 (C) (xxiii) of The GFR, 1963, one Copy of first aid leaflet issued by the Directorate general of Factories Advice Service and labour Institutes, Government of India, Mumbai shall be kept in each first aid box. 
3.	Section 111 A Safety Training records was not available.	Safety Training as required under Section 111 A (ii) shall be provided to all employees by Government

Sr. No.	Observation	Recommendation
		approved occupational safety and health training institute or DISH approved Competent Person.
4.	Record of undertaking taken from employee for availing training on Precaution, Hazard, Precautionary notice and instruction was not evident.	As per Part 2 (4) of Schedule XIX of rule 102 of the GFR, 1963 An undertaking from the workers shall be obtained within 1 month of their employment and for old workers employed within one month of coming in to operation of these rules, to the effect that they have read the contents of the cautionary notices and instruments, understood by them and would abide by them.
5.	**Safety shower**: • Location was not visible. **Acid – alkali Storage tanks**: • No cautionary board observed.	• Display circle with a green & white colour strip inside circle at such a location, which can be easily seen from a distance. Now a day there is a practice to provide green light (always ON) to identify SS location from a distance. Display of notice as below **CAUTIONARY NOTICE** **DANGER** **Acids and Alkalis cause severe burns and vapours thereof may be extremely hazardous. In case of contact, immediately flood the part effected with plenty of water for at least 15 minutes.**
6.	First aid box available in the factory but notice regarding first aid was missing.	As per Rule 70 A of The Gujarat Factories Rules (GFR), 1963, A notice containing the names of the persons working within the premises of the factory who are trained in first aid treatment and who are in charge of the first aid boxes or cupboards shall be posted in every factory at a conspicuous place and

Sr. No.	Observation	Recommendation
		near each such box or cupboard. The notice shall also indicate work room where the said person shall be available. The name of the nearest hospital and its telephone number shall also be mentioned prominently in the said Notice. **List of Trained First Aiders: -** 1. Mr. xxxxx (Mo. 00000) 2. Mr. vvvvv (Mo. 00000) **Hospital Name:** 3. XXX Hospital (Contact No.: 00000) 4. YYY Hospital (Contact No.: 00000)
7.	Siren code was missing in entire plant premises.	Siren code would be displayed in prominent manner and in local language in entire plant premises as suggested below.  
8.	SOP/Do's & Don'ts not displayed at all operation areas and Chemical storage area.	SOP/Do's & Don'ts to be displayed at all operation areas and Chemical storage area.
9.	Confined space entry was not identified and not displayed.	As per Rule 102, Schedule 19, Part II, Para (18) (1) (a) of The Gujarat Factories Rules, 1963, Confined pace entry to be identified and to be displayed.

Sr. No.	Observation	Recommendation
10.	Logbook of every entry into or work in confined spaces was not evident.	As per Rule 102, SCHEDULE 19, PART II, (18) of The Gujarat Factories Rules, 1963, The manager shall maintain a log book of every entry into or work in, confined spaces and such record shall contain the details of persons assigned for the work, the location of the work and such other details that would have a bearing on the safety and health of the persons assigned for this work. The log book so maintained shall be retained as long as the concerned workers are in service and produced to the Inspector when demanded.
11.	Maximum Permissible Working Pressure (MPWP) was not marked on Pressure Gauge (PG) in Plant.	As per Rule 61 (4) (b) of Gujarat Factories Rules 1963, MPWP shall be marked with prominent red colour on Pressure Gauge.
12.	Safe Working Load (SWL) of material hoist/lift was not displayed.	As per the **Section 28 (c) of The Factories Act 1963,** the maximum safe working load shall be plainly marked on every hoist/lift, and no load greater than such load shall be carried thereon. Testing date and due date also to be marked.
13.	Emergency escape route with arrow and symbols were not available in operation area.	Paint or fix sticker for Emergency escape route with arrow & symbols, which can be easily seen from a working area and at junction where there is more than one way- would be displayed as suggested below.

Sr. No.	Observation	Recommendation
14.	Emergency Exit signage was displayed at few locations.	As per DISH notification no. DISH/H-Safety/2019/958 to 1057 Dated: 24/09/2019. Emergency Exit signage and lighting system shall be as per IS code: 10322/12349. In case of total dark out, fluorescent stickers or indicting the way to exit to be displayed.
15.	PPE matrix display was not found in plant area & Usage of PPEs found limited	Prepare PPE matrix to use specific PPE on the bases of operation / risk / hazard. Give training to employees, and instruct to use PPE as decided in PPE matrix during the activity.
16.	Mock drill records as per Rule 68-J not evident.	As per Rule 68-J (12) (4) (5) of The Gujarat Factories Rules, 1963 The occupier shall ensure that a **mock drill** of the on-site emergency is conducted at least once in every six months and reports of the shall be made available to DISH.
17.	Factory handling hazardous substances however formal system for participation of workers in health and safety found inadequate.	As per Section 41 – G of The Factories Act, 1948 and there under Rule 68 – Y of The Gujarat Factories Rules, 1963 in every factory carrying out hazardous processes or handling hazardous

Sr. No.	Observation	Recommendation
		substances shall set-up a **Safety Committee** consisting equal numbers of representatives of workers and management.
18.	Work Permit System was evident but found inadequate.	Work Permit System (WPS) to be strengthen. WPS form to be updated and effective JSA to be performed.
19.	Trained first aider was not available	As per section 45, Para III of the Factory Act 1948, First aider should obtain certificate in first aid treatment recognized by State Government Institute.
20.	HAZOP study was not evident for process plant.	HAZOP study shall be carried out for all process plant to identify potential hazard and operability issues
21.	ERT Team/First Aider/ Fire frightening team was not available in factory.	The occupier shall arrange to have 10 % ten percent of the workers trained in the use of first aid, fire-fighting appliances and in the rendering of specific first aid measures taking- into consideration the special hazards of the particular process -– schedule 19, rule 102
22.	Factory handling hazardous substances however records of workplace monitoring was not evident.	As per the Section 7-A of The Factories Act, 1948, every occupier shall ensure to carry out **workplace monitoring** and records shall be maintained Form No. 37 as prescribed under Rule 12 – B of The Gujarat Factories Rules, 1963.
23.	Pre-employment medical check-up of employee and worker was done but certificate of fitness in Form No. 33 was not evident.	**As per Rule 68 T (2) of The Gujarat Factories Rules, 1963** No person shall be employed for the first time without a certificate of fitness in Form No. 33. certificate of fitness in Form No. 33 shall be maintained.
24.	Fire drill records was not evident.	As per Rule 66-A (12) (c) of The Gujarat Factories, 1963 **Firefighting drill** shall be held as often as necessary and at least once in every period of 2 months and records maintained.
25.	Identification no., Resistance value, last test date, next due test date was missing on earth pit.	Resistance value, last test date, next due test date should be displayed on each earth pit.

Sr. No.	Observation	Recommendation
		 Colour should be written on Earth pits. (shown in picture.)
26.	Mechanical guarding of rotating parts of machines at many places found inadequate;	As per the **Section 21 of The Factories Act, 1948** all rotating parts of machines shall be adequately guarded to restrict the accessibility while in motion. It is recommended to verify all machines for adequacy of the existing guarding and rectify on high priority.
27.	Shock treatment chart in English or Hindi and local language was not display in all panel room.	Shock treatment chart in English or Hindi and local language should be display in all panel room.
28.	Display board of hazardous waste was not found at entrance of factory.	As per supreme court order no. W.P. No. 657 of 1995 Dated: 14/10/2003, Display board of hazardous waste should be updated on regular basis. Updated board as per valid GPCB consent.

Sr. No.	Observation	Recommendation
29.	Hazard Identification & Risk Assessment (HIRA) was not evident.	As per Rule 68-0 (4) (e) of The Gujarat Factories Rules, 1963 Hazard Identification & Risk assessment (HIRA) to be prepared and regularly update at every 2 years.
30.	Hazardous Area Classification (HAC) study was not available.	As per Schedule XIX Part-III Paragraph under Rule 102 of The Gujarat factories Rules, 1963 Hazardous Area Classification shall be done and accordingly electrical equipment shall be installed.
31.	Workmen / employees handling hazardous substances were not adequately aware about the type of substances stored / handled, hazards of substances, their hazardous properties, exposure consequences, control measures, personal protective equipment to be used etc.	As per Section 41 B / C / H of The Factories Act, 1948 and Rule thereunder i.e., Rule 68-K of The Gujarat Factories Rules, 1963 The occupier of the factory involving a hazardous process shall supply information to all workers individually through **booklets** or leaflets and display of cautionary notice at workplaces.
32.	Guard to pump coupling, agitator coupling & blower or agitator belt or chain was not found in most the cases.	This is a statutory requirement to cover all rotating parts to prevent from body injury. It is recommended to provide rigid guard to cover all rotating parts and anchor to ensure that it will be in position all the times.
33.	Safety shower provided in plant area where acid / Alkyl / liquid	It is recommended to display notice as per GFR as below

Sr. No.	Observation	Recommendation
	chemicals were handled but there was not display of cautionary board found near safety shower.	CAUTIONARY NOTICE Danger Acid and Alkalis cause severe burns & vapor thereof may be extremely hazardous. In case of contact, immediately flood the effected parts with plenty of water for at least 15 minutes.
34.	Periodical medical examination was carried out and records maintained in form-32, but employees details was not complete filled and entire units employees were covered in a single register i.e. no any identification or segregation for other units like unit-2, 3, etc.	Employees adequate details to be filled in form-32. If possible separate register to be prepared for all the units or need proper identification / segregation for each unit.
35.	Illumination study records was not evident.	As per sec-17 of The Factories Act-1948 & as per rule-31 of the Gujarat Factories Rules-1963, in every part of the factory where workers are working or passing there shall be provided and maintained sufficient and suitable lighting, natural or artificial, or both. Illumination study to be carried out for adequacy of the lighting facility.
36.	QRA study was not evident	QRA study to be carried out for consequence analysis.
37.	On site emergency plan (OEP) was available but not updated.	On site emergency plan (OSEP) should be prepared as per Rule 68-J, Schedule 8 A of The Gujarat factories Rules, (GFR) 1963.
38.	Factory handling hazardous substances but Self-contained Breathing Apparatus (SCBA) set not available for rescue or handling toxic release.	It is recommended to provide SCBA set in plant to reduce the response time in case of emergency.
39.	a) Piping color-coding chart display not found in operational area and name of pipeline chemical content not marked on all pipelines, this might result wrong chemical charging or isolation. b) Flow direction arrows marking not found for all pipelines. (pipelines outer surface had sticky material)	a) There is IS 2379 for pipeline color-coding for specified pipeline content / chemical OR develop in-house colour coding norms and display in plant area and follow the same or display name of chemical on pipelines. b) Mark flow direction arrow on all pipelines, which will help for quick and correct opening or closing of valve during emergency.
40.	Over all housekeeping not to the required level, storing of chemical drums together outside the building, direct to sunlight, without second layer of contentment and	It needs review for overall improvement.

Sr. No.	Observation	Recommendation
	without study of chemical compatibility of different chemicals.	
41.	TFH (Thermic Fluid heater) testing records i.e. Oil testing report was not found.	As per the Sub-rule 18 of Rule 68 – D of The Gujarat Factories Rules, 1963 Thermic fluid shall be tested by competent person for suitability at least in every three-month period. Such test shall include test for acidity, suspended matter, ash content, viscosity and flash point.
42.	Record of Pressure test of the heater coil of thermic fluid heater was not evident.	As per rule 68 D (16) of The Gujarat Factories Rules, 1963 The heater coil shall be subjected to pressure test by competent person once at least in every 12 months. The test pressure shall not be less than twice the operating pressure. Record to be maintained.
43.	Thermic Fluid heater logbook was not evident.	Thermic Fluid heater logbook to be maintain as per rule 68 D of The Gujarat Factories Rules, 1963
44.	Factory is having huge fire load in plant and fire hydrant system, fire extinguishers available but fire load calculation / fire adequacy report not evident.	As per Rule 66 – A of The Factories Rules, 1963 **fire load calculation** shall be carried out to ensure adequacy of existing fire protection facility.
Engineering Stores		
45.	Dump of scrap • Engineering material • Metal scrap • Combustible material • Carboys, • Damaged glass wares (Hazardous waste, legal requirement not complied) • Inadequate illumination • No walkway • Hundreds of drum of mother liquor drums kept on floor, no access in case any drum leak • Unidentified compressed gas cylinder • No ventilation • No emergency escape routes identified • It's a highly unsafe condition prone for a serious incident	All such material must be segregated and disposed off / stored with clear labels as required. Unidentified compressed gas cylinder must be destroyed after necessary such cylinder scrap procedure / risk assessment
46.	Operating stores had many electrical motors, status unidentified.	All such equipment must be tagged with details for end use / scrap
47.	Contractor fab. Shop nr. stores Gas cutting set	• Inspection checklist to be prepared. • Oxygen & DA cylinder to be chained properly.

Sr. No.	Observation	Recommendation
	• Oxygen and DA cylinders were not chained • No provision of flash back arrestors at both gas cylinders and cutting nozzle sides • Oxygen gas regulator, regulator dial was found damaged	• Flash back arrestor to be provided on gas cylinders & cutting nozzle sides. • Damaged regular to be replaced.
48.	Welding machine was • Not provided with RCCB / ELCB protection for human • No electrical inspection evident	All electrical hand tools / welding machines / temporary electrical extension boards must be provided with RCCB to protect human from shock hazard.
49.	There were more than 100 drums full of chemical • Not identified • No identification of stored location • No shed, stored in sun • Few drums were found bulged due to pressure inside • No secondary containment to control leak / spill	All such drum storages, elsewhere also, must be identified, have secondary containment, shed. Any spill / leak may lead to spreading of chemicals in plant due to drain channel nearby. Mother liquor drums stored in shed must be accessible to meet any emergency actions,
50.	Hydrogen peroxide carboys, isopropyl containers were also kept without shed. No containment.	High potential fire and chemical reactive chemicals. Store as per MSDS requirement. Recommended storage is to keep below 25 deg c
51.	RV drier: No operating instructions displayed	SOPs must be prepared. This true for all areas SOP preparation.
52.	The empty drums were cleaned and sold to the customers. There was no proper decontamination skid. No labelling as DECONTAMINATED was fixed on cleaned drums	These practices must be followed. Any drums sold must be certified as decontaminated with details of decom.
PESO Storage Area		
53.	Toluene and methanol were stored in the area in u/g tanks • There was no display of PESO licence validity • The area was uneven, surrounded by lot of other material. • Pump start / stop push buttons were not easily accessible. • Tank LI was removed for maintenance and the nozzle was open to atmosphere,	It's a government licensed area and must be maintained as per legal requirement.

Sr. No.	Observation	Recommendation
	exhausting flammable vapour in the area • No fencing around the area • Few vents were not provided with flame arrestor • Unloading pump motor coupling guards were not as required. • Motor had no two earthings • Few plugs not plugged in pumping system (Vents / drains) Flame arrestors were found at few locations (PM not evident)	
54.	One tanker of methanol GJ-06-AU-4455 was ready for unloading • Correct name of chemicals and UN no. written on tanker were changed manually by unauthorised person from previous chemicals. • A tanker check list before entering the gate by security not evident • Tanker reversing in case of any emergency or after unloading is not recommended in plant	• Such tempering must be avoided. • The check list must include all such points for safe movement and smooth unloading • Review for provision of forward movement during such conditions
55.	Tanker unloading static discharging system was not reliable. In case of earthing failure, there was no indication. Unloading would continue without any static discharge provision. Highly unsafe condition. Plastic carboys and funnels were used during tanker unloading to handle draining etc.	It is recommended to provide smart earthing system which will give alarm and discontinue unloading in case of earthing failure. Any hydrocarbon collection container must be metallic including funnel and these must be connected to earthing for static discharge.
Process Plant		
56.	Hydrant hose box in front of plant entry • Was not identified • No inspection evident	Fire critical equipment must be maintained by periodical inspections.
57.	FEs were available in the area, need to be managed.	• To be numbered • Mounted at a height of 750 mm (Bottom of FE) • Periodical check • Easily visible from a distance

Sr. No.	Observation	Recommendation
		• Back side with zebra painting for easy locating • Travel distance not more than 23 meters from any location
58.	One fork list was moving with a drum put on fork (No pallet), may slip / fall.	This is highly unsafe act and must be avoided
59.	Reactor body and connecting piping static earthing was not visible.	All equipment in plant vessels / piping / storage tanks / day tanks must be provided with double earthing and connected to separate earthing network to avoid any spark and fire incident.
60.	No fire detector / heat detectors / smoke detectors in plant for pre fire alarm.	Review for installation of such fire alarm system for timely responding the emergency
61.	Toluene tank / Cummins tank were not provided with • Overflow / High level / low level protection device • Vent • Level indication device	Any error leading to overflow may lead to a high potential fire incident. Review for provision of such critical operating devices.
62.	For all chemical day tanks / storage tanks / equipment had no periodical inspection schedule for healthiness of equipment.	Such periodical inspections must be planned and carried out for safe environment.
63.	GL-8 JB was a flame proof equipment but its plug was open, not fulfilling the purpose Agitator cover box, no screws Only 2 / 4	All such electrical JBs and installation maintained
64.	Exit route in plant were adequate to rush to the safe location.	Adequate displays must be provided for easy and safe escape. To be fluorescent to be visible during night hours also.
65.	Line identification / colour coding, not followed.	All lines must be identified with service and flow direction for easy and safe operating
66.	Caustic tank • No adequate dyke volume • No overflow protection • No chemical proof tiling on floor • No LI / high / low level protection • No identification / capacity / safe height / service / NFPA diamond display • No periodical inspection schedule • No provision of safety shower and statutory display for emergency actions	All such hazardous chemical storages must be maintained to avoid and emergency. Statutory warning as below to be displayed Cautionary Notice - Danger Acids and Alkalis cause severe burns and vapours thereof may be extremely hazardous. In case of contact, immediately flood the part effected with plenty of water for at least 15 minutes.

Sr. No.	Observation	Recommendation
	• No operating platform for safe movement of operator, fall hazard • No use of PPE display	
67.	H_2SO_4 storage tank In addition to above observations, following to be done • Dyke integrity not maintained • Structure stability not checked	As above to be complied
68.	One ordinary electrical extension board was in use without any ELCB / RCCB protection, near entry gate.	No ordinary equipment / electrical devices must be used. These are source of fire and defeat purpose of area classification
69.	Centrifuge for processing Cover was in open condition when centrifuge was in operation No protections (viz no start without cover in close condition, stopping and breaking if cover opened in running condition)	It is a hazardous equipment and its protections must be in line and periodically checked for functioning.
70.	Oil circulation pump area It is adjacent to area where flammable material was used viz methanol. IPA etc. The area had all non-flame proof equipment / installation	Review for considering it as per area classification considered for main area.
TFH / Electrical Panel		
71.	Thermic fluid unit • No record of shutdown evident • No periodical analysis of thermic fluid • No interlock protections checking • No tube hydro test data available (1.5 times working pressure) • No equipment body earthing visible	The equipment must be maintained as per requirement of GFR 68D
72.	Illumination was not adequate in the operating area	To be surveyed and maintained
73.	Electrical panels Well maintained viz Numbering and description of panel Clean and well places rubber mats	Good practice
74.	W5E-04 Equipment structure had badly corroded.	To be repaired to avoid collapsing the structure when in operation
R&D Laboratory		

Sr. No.	Observation	Recommendation
75.	No detector viz smoke / fire / heat.	Review for provision of such detectors for emergency quick response.
76.	Spill management procedure not available	To be prepared for spill of fire / toxic material handling
77.	Compatibility chart not displayed.	To be prepared and displayed for the chemicals used
DG set		
78.	Diesel tank • Equipment identification not done • No earthing to tank • Level indicator Opaque • O/L valve handle was missing	To be attended
200 KVA Transformer		
79.	• Fencing / fencing gate earthing not visible • EP number missing • FE not visible • Discoloured silica gel	To be done
RM / FG Stores		
80.	Battery charger inside the stores. It is a source of fire due to liberation hydrogen • No instructions displayed • No use of PPE displayed • Containment around battery	To be relocated to well-ventilated place.
81.	OH Crane A locally fabricated cage was in use for lifting and shifting of material. These were not identified with number, no evidence of inspection. No braking system	All such lifting items must be checked periodically for corrosion, crack on weld joint of lifting rods, other defects to ensure its integrity.
82.	Main rack • Rack not displayed with capacity • Few locations found damaged due to hit by fork lift • Damaged pallets in stored • Drums kept on damaged pallet, a danger of falling • No adequate ventilation • No adequate illumination • No containment plan in case of spillage of chemicals	Capacity to be displayed To be provided with hard barricading to avoid direct hit A plan to inspect / reject pallets
Fire Water Pumping System		
83.	Pump was kept in vehicle parking, surrounded by two wheelers • Access was difficult.	It is an emergency equipment and must be maintained. Level monitoring to be done at least shift wise

Sr. No.	Observation	Recommendation
	• No details of pump capacity / reservoir capacity / start / stop philosophy available • Process water also from the same reservoir (No reserve for fire system) • Reservoir level monitoring not evident • Pump discharge was not connected with fire water net work	Pump discharge line must be connected permanently instead of connecting with hose (Present practice)

9. Occupational Health & Safety Audit as per IS 14489: 2018

9.1 ANNEXURE – B: Examination of Records (IS 14489: 2018)

Observations including nonconformities and good practices by the auditee factory are given below in Annexure B to C. Necessary recommendations / suggestions or improvement measures are also suggested in the last column.

Sr. No.	List Of Records / Documents	Pertaining Legal Requirement	Current Status / Observation
1.	OH & S policy	The GFR- Rule 12-(C), 68- O.	Quality policy available in the Organisation.
2.	Safety organization chart	The FA 1948 – Section 40- B.	Safety Organisation Chart available in the Organisation.
3.	Training records on safety fire and first-aid	The FA 1948 - 7 (A), 111 (A) The GFR- Rule GFR – 12 (C) 4(g).	Training records on safety fire and first-aid available in organization.
4.	Record of plant safety inspections	The GFR- Rule 112.	Informal records of safety inspections available in organization.
5.	Accident investigation reports	The GFR- Rule 103, 111.	Accident investigation reports was not evident.
6.	Accidents, dangerous occurrences and near miss incidents - statistics and analysis	The GFR- Rule 103, 111.	Accidents, dangerous occurrences and near miss incidents - statistics and analysis were not evident.
7.	Record of tests and examinations of equipment and structures as per statutes	The GFR- Rule 58, 59, 60,102.	Form No. – 10, 11 and stability certificate available.

Sr. No.	List Of Records / Documents	Pertaining Legal Requirement	Current Status / Observation
8.	Standard Operating Procedures (SOP) for various operations	---	Standard operating procedures were available for various operations.
9.	Record of work permits;	The GFR- Rule 102 Chemical works sch. 19.	Record of work permits were available. **List of work permits:** 1. Hot work 2. Confined Space 3. Height Work 4. General Work
10.	Record of work environment monitoring (Flammable, toxic and explosive substances)	The GFR- Rule 68 J and 12 – B The FA 1948 Section 7-A (2) (E).	Record of work environment monitoring (Flammable, toxic and explosive substances) was available informally.
11.	Maintenance, testing and calibration records of fire detection and firefighting equipment	The GFR- Rule – 66A.	Maintenance, testing and calibration records of firefighting equipment was available.
12.	Medical records of employees	The GFR- Rule 68 R.	Medical records of employees (Form No. 32) were available.
13.	Records of industrial hygiene surveys (noise, ventilation, illumination, dust etc.)	The GFR- Rule - Rule 12-B The FA 1948 - Section 7A (2)(E).	Records of industrial hygiene surveys (noise) was available. Illumination, dust to be carried out.
14.	Material Safety Data sheets	The GFR- Rule 68 J, MSIHC Rule 1989, Rule 17, Schedule 9.	MSDS of All chemicals, RM, Intermediates and Product were available.
15.	On-site emergency plans and record of Mock Drills	CA EPPR 1996, The GFR- Rule 68 J, The FA 1948 Schedule – 8-A.	On-site emergency plan was available. Record of Mock Drills was available. Last mock drill carried out: - 12/08/2023
16.	Records of storage of hazardous solid waste and its disposal	The GFR- Rule - 18, The FA 1948 section 12.	Records of storage of hazardous solid waste and its disposal was available.
17.	Records of gaseous emissions and effluent discharges to the environment	The GFR- Rule - 18, The FA 1948 section 12.	Records of gaseous emissions and effluent discharges to the environment was available.
18.	Housekeeping inspection records	The GFR – Rule 16, 17, 17 A, The FA 1948 section 11.	Housekeeping inspection records was available.
19.	Minutes of safety committee meetings	The GFR- Rule – 68 F, Y.	Safety committee under formation
20.	Statutory licenses and approvals	The GFR- Rule -3.	Statutory licenses and approvals were available.

Sr. No.	List Of Records / Documents	Pertaining Legal Requirement	Current Status / Observation
21.	Records of any modifications carried out in plant or process	The GFR- Rule -3,7.	Records of any modifications carried out in plant or process was available informally.
22.	Maintenance procedure and records	---	Preventive Maintenance procedure and records were available.
23.	Instrumentation and equipment calibration and testing records	The FA 1948, Section 36.	Instrumentation and equipment calibration and testing records available at technical department
24.	Planned shutdown maintenance procedures	---	Procedure for planned shut down was available.
25.	In service inspection manuals, records including that of material handling	---	All material handling equipment service inspection records, manuals available.
26.	OH & S budget	The GFR- Rule 12 C & 68-0.	OH & S budget available as & when required.
27.	Inspection books and other statutory records;	The GFR- Rule – 112.	Inspection books and other statutory records were available.
28.	Records of previous audits and safety analysis;	The GFR- Rule 68 J	This is a first safety audit.
29.	Procedures for safe transportation of hazardous substances;	MV Act.	Procedures for safe transportation of hazardous substances was available.
30.	Calibration records;	---	Instrumentation and equipment calibration and testing records available at technical department.
31.	Records for breakdown of plants during the process of manufacture;	---	Records for breakdown of plants during the process of manufacture was available at technical department.
32.	Records for waste material generated and their disposal;	---	Records for waste material generated and their disposal record was available.
33.	SOP for disposal of waste materials; and	---	SOP for disposal of waste materials was available.
34.	Records for issue of PPE items to the personnel working in process building.	---	Records for issue of PPE items to the personnel working in process building was available at safety department.

9.2 ANNEXURE – C: General Points (IS 14489: 2018)

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	C-1 OH & S Management		
	C-1.1 OH & S Policy (Rule 12-C And / Or 68-O of The GFR 1963)		
1.	Does the organization have OH & S Policy?	Quality policy was available. OH&S policy was under preparation.	---
2.	Who has signed the OH & S policy?	OH&S policy was under preparation.	84. As per rule 68-O of the GFR-1963, OH&S policy to be signed by occupier
3.	Whether the OH & S policy is per guidelines of the statutory provisions?	OH&S policy was under preparation.	---
4.	When was the OH & S policy declared and adopted?	OH&S policy was under preparation.	---
5.	Whether the OH & S policy reviewed periodically?	OH&S policy was under preparation.	85. As per rule 68-O of the GFR-1963, OH&S policy to be reviewed periodically.
6.	Whether the OH & S policy is available in local language and made known to all?	OH&S policy was under preparation.	86. As per rule 68-O of the GFR-1963, OH&S policy to be made available in a language understood by the majority of workers and explained to them.
7.	What was the last date of updation?	OH&S policy was under preparation.	---
	C-2 OH & S Organizational Set Up		
	C-2.1 Safety Department - The FA Section 40-B		
8.	Does the factory have a safety department and what is strength of safety department?	Factory has a safety department and strength of Safety department was 1 person.	---
9.	Whether the strength and qualifications of Safety Officers are as per the statutes?	We observed that, strength and qualifications of Safety Officers were as per the statutes Notification No. **KHR-529-FAC-1077-36785-M-(1)** dated 19.12.1985	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
10.	Does the head of safety department report to the Chief Executive?	The EHS manager report to Unit head & Director.	---
11.	How often are the safety officers retrained in the latest techniques of total safety management? What is the frequency of retraining?	Safety officer trained in periodic refresher courses & participated in various activities taken up by the Factory inspectorate and local agencies	---
12.	What additional duties the safety officer is required to do?	Only Safety related Functions.	---
13.	What is the power of safety officer vis-a-vis unsafe condition or unsafe act?	The power of safety officer is to stop any dangerous operation. If there are any unsafe conditions or acts, these will be brought to the notice of the concerned HOD and in the case of imminent danger, the job stopped until unsafe conditions / act was removed / corrected.	---
	C-2.2 Safety Committee(s) (The FA Section 41-G, Rule 68-F, Y of the GFR 1963)		
14.	Does the factory has a safety committee(s)? What are the types, structures and terms of reference of the committees?	Safety Committee was under preparation.	87. As per rule 68-Y of the GFR-1963, safety committee to be prepared and at regular interval committee meetings to be carried out.
15.	Is the constitution of the safety committee(s) as per the statute?	Safety Committee was under preparation.	---
16.	How are the members of safety committee(s) selected? (elected / nominated)	Safety Committee was under preparation.	---
17.	How often are the meetings of safety committee(s) held?	Safety Committee was under preparation.	88. As per rule 68-Y of the GFR-1963, safety committee to be prepared and at once in a quarter committee meeting to be carried out.
18.	Are the recommendations of the committees(s) implemented?	Safety Committee was under preparation.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
19.	Are the minutes of the safety committee(s) meetings circulated among the members?	Safety Committee was under preparation.	89. MOM of safety committee to be circulated to members.
20.	Are the minutes forwarded to the trade union(s) and chief executive and occupier?	Safety Committee was under preparation.	90. MOM of safety committee to be forwarded to chief executive & occupier.
21.	Whether the management and trade union play their active roles in supporting and accepting the committee(s) recommendations?	Safety Committee was under preparation.	---
22.	How are the safety committee(s) members apprised of the latest development in safety, health and environment?	Safety Committee was under preparation.	91. Members of the safety committee's to be sent to participate in external safety training program i.e., at DGFASLI, NSC, GSC, GLI, etc.
	C-2.3 Safety Budget (The GFR- Rule 12-C & 68-0)		
23.	What is the annual safety budget?	EHS Budget was available as & when required.	---
24.	How much percentage is this budget of the total turnover of the company?	Adequate - made available as and when required.	---
25.	How much budget has been utilized till date?	Data available at EHS department.	---
26.	Is the safety budget adequate?	It's informed that Safety budget was adequate.	---
27.	How is the safety budget arrived at?	Based on the previous year's utilization & present requirements.	---
28.	What is the pattern of expenditure for the last five years?	Mostly Increasing trend. Price hike & the requirement.	---
29.	What are the approved sanctions for the expenditure in this budget?	Data available with EHS department	---
30.	Does this budget get reflected in the annual report of the company?	Safety budget was not reflected in the annual report of the company.	---
	C-3 Safety Manual (The FA Section 41-B,41-C,41-H, GFR-Rule 68-K)		
31.	What is the periodicity of updation / review of safety manual?	Safety booklet was not evident.	92. Safety booklet to be prepared and made known to all.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
32.	Does the safety manual adequately address all the hazards in the plant?	Safety booklet was not evident.	---
33.	Are the employee made aware of safety rules?	Safety booklet was not evident.	---
	C-4 Standard Operating Procedures (SOP)		
34.	Are written Standard / safe operating procedures available for all operations and processes?	Standard operating procedures were available.	---
35.	Whether the written Standard / safe operating procedures are displayed or made available and explained in the local language to the workers?	The written Standard / safe operating procedures were displayed or made available in the english only.	93. Standard / safe operating procedures should be made available in local or national language and displayed it at conspicuous places.
36.	Whether concerned section and safety department prepares standard / safe operating procedure jointly?	Concerned section and safety department prepares standard / safe operating procedure jointly.	---
37.	Are standard / safe operating procedures reviewed and updated?	Standard / safe operating procedures reviewed and updated by concerned section as and when required.	94. SOP to be reviewed after any change in process and Plant equipment.
38.	Have the workers been informed of the consequences of failure to observe the standard /safe operating procedures?	Workers were informed about the consequences of failure to observe the standard / safe operating procedures during training.	---
	C-5 Plant Modification Procedures		
39.	What is the system for effecting any change in the existing plant, equipment, or process?	MOC (Management of Change) system for controlling any change in the existing plant, equipment or process was not evident.	95. MOC / Change control procedure to be prepared for any change in plant, equipment or process.
40.	Whether the P & I diagram, and other related documents are updated accordingly?	Its inform that, P & I Diagram, and other related documents were updated.	---
41.	Whether hazard assessment done before implementation of modification?	MOC (Management of Change) system was not evident.	---
	C-6 Work Permit System (The GFR- Rule 102 Chemical Works Sch. 19)		

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
42.	What types of work permits exist in the factory?	Record of work permits were available. **List of work permits:** • Hot work • Confined Space • Height Work • General Work	----
43.	Are the necessary forms detailing required safety precautions have been prepared and used for each type of work-permit?	Necessary forms detailing required safety precautions have been prepared and used for each type of work-permit.	---
44.	Is the responsibility assigned to authorized person for issuing of safety work permit?	Responsibility assigned to authorized person for issuing of safety work permit	---
45.	Is the copy of safe work permit sent to safety officer before execution of the job?	One copy of work Permit sent to safety Officer before execution of the job.	---
46.	Is validity period specified in the safety work permit?	Date, time and location, validity was mentioned in the safety work permit.	---
47.	Are the records of work permit available and maintained in proper order?	Records of work permit was available.	---
	C- 6.1 Control Measures for Work at Height		
48.	Is adequate safe access provided to all places where workers need to work?	Adequate access provided where workers need to work.	---
49.	Are all such access in good condition?	Most of access was found in good condition.	---
50.	Are all scaffolds are properly designed and erected?	Evident of scaffolding uses was not found. As & when required, erected through contractor only.	96. When scaffolding installed, its guideline should be followed
51.	Are scaffolds inspected every day before work begins?	Evident of scaffolding uses was not found.	---
52.	Are ladders securely clamped or lashed in place?	Evident of scaffolding uses was not found.	---
53.	Are planks in good condition?	Evident of scaffolding uses was not found.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
54.	Are scaffold walkaways, platforms, runs or stairs free of debris, grease, any unnecessary obstruction and projecting nails?	Evident of scaffolding uses was not found.	---
55.	Are the scaffolds higher than 20 m.? If yes, is netting or intermediate railing provided between toe-boards and hand railings?	No scaffolds higher than 20 m. Third party hired for above 20 m height work.	---
56.	Are folding stepladders properly used?	Folding stepladders used properly.	---
57.	Are ladders set up at the proper slope of about 1:4?	Ladders set up proper slope of about 1:4 Ratio.	---
58.	Do workers use hand lines to lift tools or materials?	Workers used hand lines to lift tools & materials.	---
59.	Are proper ladders used around electrical hazards?	Proper ladders used around electrical hazards was not evident.	97. Special ladders should be used around electrical hazards.
60.	On sloping roofs, are crawling boards, lifelines, safety belts and edge protection provided where needed?	Safety belt and edge protection was provided where necessary.	---
61.	Whether the weak spots, skylights, or deteriorated asbestos-cement boards through which a worker might fall while working in the roof has been identified and safety net provided appropriately?	Week spot, skylights, or deteriorated asbestos-cement boards was not seen.	---
62.	Are the workers being medically examined for their fitness to work at height?	The workers medically examined for their fitness to work at height was not evident.	98. Medical examination records to be made available for fitness to work at height.
	C-6.2 Work in Confined Space (The FA Section 36, GFR-Rule 102 Schedule 19)		
63.	Is work permit system followed for working in confined space?	Confined space permit system followed for working in confined space.	99. As per Rule 102, Schedule 19, Part II, Para (18) (1) (a) of The Gujarat Factories Rules, 1963, Confined pace entry to be identified and to be displayed.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
			100. Confined space entry to be done with competent person authorization only. 101. Kindly refer Chapter-8 "Annexure-A Area of Improvements"
64.	Whether monitoring of the atmosphere inside the confined space is carried out and ensured that there is no flammable or toxic gas in the area?	Monitoring of the atmosphere inside the confined space was carried out. (O_2, LEL, etc. checked)	102. Confined space entry to be monitored by DISH approved competent person only.
65.	Whether the person entering the confined space is using suitable personal protective equipment (PPE)?	The person entering the confined space used all required personal protective equipment.	---
66.	Is rescue team available in case of any emergency?	Team available immediately in case of any emergency.	---
	C-7 Contractors' Safety System		
67.	Is there any system for selection of contractors?	Contractors were selected upon fulfilling the pre-qualification criteria.	---
68.	Are there any guidelines on contractor's safety and training?	Guidelines on contractor's safety and training record was not evident.	103. Guidelines on contractor's safety and training to be made.
69.	Whether contract document includes necessary safety and welfare clauses as per statutes?	Yes, necessary documents were available.	---
70.	Is there any programme to ensure use of PPE by contractor's personnel?	Training, supervision and site monitoring to ensure use of PPE by contractors' personnel.	---
71.	Do the contractors have their own safety organization?	They were trained & experienced. They co-ordinate with factory Safety Dept.	---
72.	Are the contractors reporting all accidents and injuries?	The contractors reporting accidents and injuries.	---
73.	Are contractor workers trained to observe safety at workplace?	Contractor workers trained by safety officer to observe safety at workplace.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
74.	Whether contractor workers are engaged in process / operations? If yes, are they aware of safe operating procedures?	Contractor workers were engaged in process / operations. Contractor workers were aware of safe operating procedures by training.	---
	C-8 Plant Design and Layout (The FA Section 87, GFR-Rule 102)		
75.	Whether hazardous operations in the plant are segregated?	Hazardous operations in the plant were segregated.	---
76.	Whether occupational health & safety aspects are considered during the design?	Occupational health & safety aspects were considered during the design.	---
77.	Are all the equipment provided with adequate space for working, maintenance etc.?	The equipment provided with adequate space for working, maintenance, etc.	---
78.	Are the storage tanks provided with enough space Clearance between them?	The storage tanks provided with enough space clearance between them.	---
79.	Whether the plant layout has taken care of the movement of firefighting equipment and emergency exits?	The plant layout has taken care of the movement of firefighting equipment and emergency exits.	---
	C-9 Medical Management of Accidents (The FA Section 45, GFR-Rule 68-U,68-V,71)		
80.	Are medical facilities available with trained first aid staff and equipment in round the clock shift for all including contractors?	Occupational health centre: available FMO - Dr. Ketan Patel (Part-time)	104. OHC to be reviewed as per The Rule – 68 U of The Gujarat Factories Rule 1963. 105. First aid training to be provided from govt. approved training institute
81.	Is the ambulance van available for round the clock basis with the dedicated driver?	Dedicated vehicle was available round the clock basis with dedicated driver.	---
82.	Is there any mutual aid scheme available with the nearest hospitals to manage and treat injuries during emergency?	Yes, mutual aid scheme available with nearby Hospital – Jayaben Modi Hospital, Ankleshwar.	---
83.	Are the workers / contractor workers aware of emergency medical	Workers / contractor worker aware about the emergency medical	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	facilities?	facilities.	
	C-10 Management of Emergencies (Natural / Man-Made)		
84.	Does the system exist to detect and control these Emergencies?	Yes, Disaster Management Plan (OSEP) plan was in place. OSEP available for detect and control emergencies.	---
85.	Are the employees aware of the measures to be taken during emergencies?	Workers found aware about the measure to be taken during emergencies. Regular basis training given.	---
	C-11 Employees Selection and Placement		
86.	Whether norms are available for selection of different category of employees?	Job Description criteria available with HR Department for selection of different category of employees.	---
87.	Whether pre-employment medical examination is being conducted for employees?	Pre-employment medical examination records was not evident.	106. Pre-employment medical examination to be carried out and records to be maintained in form-33 as per rule 68-T of the GFR-1963.
88.	Is there any procedure to evaluate safety awareness and record of the employees during their promotion?	Informal system	107. Evaluation records to be maintained.
	C-12 Safety Culture		
	C-12.1 Attitudes of Managers		
89.	Do the managers follow the plant safety rules at all times?	Managers follow the plant safety rules.	---
90.	What are their attitudes towards safety reviews and audits?	Positive attitudes towards safety reviews and audits.	---
91.	What is the response of management to safety violation?	Verbal Warning & penalty provisions available.	---
92.	Whether safety related decisions are taken in consultation with the workers?	We came to know that critical safety related decisions were taken in consultation with the workers.	---
93.	What is the attitude of the	Verbal warning given for	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	managers towards non-use of personal protective equipment?	non-use of PPE	
	C-12.2 Attitudes of Workers		
94.	Whether workers are aware of the consequences of their wrong actions?	Workers were informed during training / plant round about the consequences of their wrong actions.	---
95.	Are laid down safe working procedures followed strictly?	Safe working procedures followed.	---
96.	What is the attitude of the workers towards their own mistake, which can prejudice safety?	We came to know that workers follow safety rules and procedures. If deviation observed by anybody they will accept and correct it.	---
97.	Do the workers report near miss incidents and suggest safety improvements?	Near miss incidents reporting system was not evident.	108. Training to employees for reporting near miss / unsafe condition on their work places to be done.
98.	Are the workers aware of the system of rewards and sanctions relating to safety matters?	Yes	---
99.	What is the attitude of workers towards use of personal protective equipment?	Mostly complied with PPE	---
	C-13 Statutory Licenses, Approvals and Records		
100.	Whether all the safety related Acts / Rules (with latest amendments) applicable to your organization identified, informed to all employees and complied?	Legal register with all applicable Acts / Rules identified, maintained, amended time to time and communicated with all concerned. Kindly refer Chapter-7 “Applicable Laws and Statutory Requirements” & Chapter-9, “Annexure-B: Examination of Records (IS 14489:2018)”	---
101.	Whether the licenses have been validated?	All licenses found in valid status. For Details kindly refer “Annexure-1 List of Licenses/ Certification and	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
		its validity “.	
	C-14 Motivational and Promotional Measures for OH & S		
102.	Does the factory have occupational health and safety suggestion scheme?	Safety suggestion scheme was not evident.	109. Safety suggestion scheme to be implemented.
103.	Are occupational health and safety contests organized in the factory?	No	110. Occupational health and safety contests should be organized in the factory during safety day celebration.
104.	Does the factory participate in National Awards?	No	---
105.	Has the factory been awarded during last five years?	No	---
106.	Does the organization publish safety bulletin /Newsletters?	Other industry’s newsletter sent to employees.	---
107.	Whether the safety bulletins are widely distributed?	Displayed on notice board	---
108.	How is the occupational health and safety information including accident statistics disseminated in the factory? (Bulletin boards, Newsletter etc.)	Notice Board	---
109.	What are the activities conducted during National Safety Day / week?	National safety day / week was not celebrated in the organization.	111. National safety day / week to be celebrated with different activities in the factory to increase awareness among workers
110.	What is the percentage of Workers participating in the various safety promotional activities?	National safety day / week was not celebrated in the organization.	---
	C-15 Hazard Identification and Job Safety Analysis		
111.	Was an initial process hazard analysis (PHA) completed?	PHA study not evident.	112. Appropriate PHA to be carried out.
112.	What are the stages of PHA? Whether a dedicated group is identified for PHA?	Dedicated group was not identified for PHA.	113. Dedicated group to be identified.
113.	Was the PHA appropriate for the complexity of the	PHA study not evident.	114. Appropriate PHA to be carried out.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	process and identify, evaluate, and control the hazards involved in the process?		
114.	Does the hazard evaluation use one or more of the following PHA methodologies: What-If Analysis, Process Checklist, Hazard and Operability Study (HAZOP), Failure Mode and Effects Criticality Analysis (FMECA), Fault Tree Analysis (FTA) or any other appropriate equivalent methodology?	PHA study not evident.	115. Appropriate PHA to be carried out.
115.	Does PHA assure addressing issues of inherent safety features with respect to material and their properties?	PHA study not evident.	---
116.	Does the PHA address the hazard identification, incidents history, consequences of failures (engineering and administrative controls), human factors, consequent analysis with respect to possible safety and health effects of failure of controls?	QRA study was not carried out	116. QRA study to be carried out.
117.	What are the stages of PHA? Whether a dedicated group is identified for PHA?	PHA study not evident.	117. PHA study not evident.
118.	Does the system exist to promptly address findings and recommendations of PHA?	PHA study not evident.	118. PHA study not evident.
119.	Are the PHA's updated and revalidated at least every five years by a qualified team to assure that the PHA is consistent with the current process?	PHA study not evident.	119. PHA study not evident.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
120.	Whether the activities requiring Job Safety Analysis have been identified?	JSA was not evident.	120. JSA to be carried out and records to be maintained.
121.	Whether the identified jobs for Hazard Identification have been carried out by trained and experienced persons?	Yes, informal hazards identification was carried out by trained and experienced persons.	121. Formal HIRA to be carried out. 122. Persons shall be trained & retrained on regular basis.
122.	Whether the checklists have been prepared on each Job Safety Analysis and are being used while carrying out the job?	JSA was not evident.	123. Checklist to be prepared for job safety analysis.
	C-16 Product Safety (Dangerous Goods {Classification, Packaging and Labelling} Rule, 2013)		
123.	Whether hazards arising from use of the products are identified?	Hazards arising from use of the products were identified.	---
124.	Whether material safety data sheet prepared for the products?	Material safety data sheet prepared for the products.	---
125.	Are all the products labelled and packed appropriately?	All the products labelled and packed appropriately.	---
126.	Whether safety instructions are given along with products?	Mandatory safety instructions including TREM Card / MSDS etc. are given along with products.	---
127.	Whether policy exists for recall of products?	Process was available for recall of products.	---
	C-17 Safety Training (The FA 1948-Sec 111(A), 7 (A))		
128.	Whether training needs have been identified?	Informal system	---
129.	Is there any programme of induction training, its duration and topics covered?	Induction training was provided to the employees / workers informally.	124. Induction training to be provided to workers. 125. Checklist for safety induction training to be prepared.
130.	Whether the assessment of the trainees has been carried out?	Verbal assessment carried out.	126. Training and assessment records to be maintained.
131.	What are the infra-structural facilities available for training?	Conference room with audio-video support was available.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
132.	Whether training is conducted by qualified person?	Training was provided internally.	127. Safety Training as required under Section 111 A (ii) to be provided to all employees by Government approved occupational safety and health training institute. 128. Kindly refer Chapter-8 "Annexure-A Area of Improvements"
133.	Whether trainers are being re-trained from time to time?	Trainers were being re-trained from time to time by internally.	---
134.	Whether proper records of training program conducted are maintained?	Training given internally.	129. Training records to be maintained.
135.	How training programs are evaluated?	Verbally	---
136.	Whether schedule for training on occupational health and safety is available and maintained?	Yes, informally	---
137.	Whether the training programmes are reviewed?	No	130. Training programmes to be reviewed
138.	Are all the employees periodically trained / retrained and what is the frequency of such training?	The employees were periodically trained.	131. At regular frequency refresher training to be carried out.
139.	Are the retraining needs identified whenever a new process / products and change in existing process introduced?	Yes, informally	132. Retraining shall be organized whenever new hazards / process changes introduced.
140.	Whether training covers top management?	Training covers employees of factory and top management.	---
141.	How many hours of safety training is given to different employees?	Data available with concern department.	---
	C-18 Change Management		
	C-18.1 Management of Change		
142.	Are there written procedures for managing change to process chemicals, technology, equipment and procedures	Informal procedure for management of change.	133. MOC (Management of Change) Procedure to be made available.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	and changes to facilities that affect the plant process / system operation?		
143.	Do the procedures assure that the technical basis for the proposed change addressed prior to any change?	Informal system for management of change.	134. MOC (Management of Change) Procedure to be reviewed, implemented and record to be maintained for all changes.
144.	Do the procedures assure that the impact of the change on safety and health addressed prior to any change?	Informal system for management of change.	---
145.	Do the procedures assure that modifications to operating procedures are addressed prior to any change?	Informal system for management of change.	---
146.	Do the procedures assure that the necessary time period for the change is addressed prior to any change?	Informal system for management of change.	---
147.	Do the procedures assure that the authorization requirements for the proposed change are addressed prior to any change?	Informal system for management of change.	---
148.	Are employees involved in operating a process, and maintenance and contract employees whose job tasks will be affected by change informed of, and trained in, the change prior to the start-up of process or affected part of process / operations?	Informal system for management of change.	---
149.	Is the safety information is reviewed and updated on changes?	Informal system for management of change.	---
150.	Are the operating procedures or practices updated?	Yes, Operating procedures / practices updated.	---
	C-18.2 Mechanical Integrity		

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
151.	Does the mechanical integrity program include for all mechanical equipment including pressure vessels and storage tanks, piping and components, relief devices and vent systems, emergency shutdown systems, pumps, control systems?	Mechanical equipment including pressure vessels and storage tanks, piping and components, relief devices and vent systems, emergency shutdown systems, pumps, control systems included in mechanical integrity program. Testing done as per norms. Form No. 10 & 11 and Stability of the Building available and valid.	135. Equipment / instrument inspection to be regularly done.
152.	Are there written procedures to maintain the on-going integrity of process equipment?	Schedule available for preventive maintenance.	---
153.	Whether training been provided to each employee involved in maintaining the on-going integrity of process equipment?	Informal training was imparted to employees on PM of equipment.	---
154.	Are inspections and tests performed on each item of process equipment included in the program?	Examination & inspection was carried out of process equipment.	---
155.	Does the inspection and test frequencies meet the manufacturer's recommendation and good engineering practice?	Inspection and test frequencies meet the manufacturer's recommendation and good engineering practice.	---
156.	Are inspections and tests performed more frequently if determined necessary by operating experience?	Inspections and tests performed more frequently when determined necessary by operating experience.	---
157.	Are deficiencies in equipment that are outside limits corrected before further use so as to assure safe operation?	Deficiencies in equipment that are outside limits corrected before further use so as to assure safe operation.	136. Ensure that equipment calibration, deviation, equipment change record to be maintained.
158.	In the construction of new plants and equipment, whether quality assurance programme is implemented to ensure that equipment fabricated	Quality assurance programme was available for construction of new plant, equipment.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	is suitable for the process?		
159.	Are appropriate checks and inspections made during equipment installation stage?	Qualified and experienced person check and supervised the equipment installation in the plant.	---
160.	Are the suitability of maintenance materials, spare parts and equipment ensured during maintenance?	The suitability of maintenance materials, spare parts and equipment ensured during maintenance.	---
	C-19 Physical Hazard		
	C-19.1 Housekeeping (The GFR – Rule 16, 17, 17 A, The FA 1948 section 11.)		
161.	Are all the passages, floors and the stairways in good condition?	The passages, floors and the stairways in good condition at most of place.	---
162.	Is glass door taped or otherwise marked to make it visible to workers?	Glass door was taped / marked.	---
163.	Do you have the system to deal with the spillage?	Dry sand available to deal with the spillage.	137. Spill kit to be provided.
164.	Do you have sufficient disposable bins clearly marked and whether these are suitably located? Are containers of refuse (waste) and trash emptied at the end of every day or soon after they are full? Are the containers or bins regularly cleaned?	Sufficient disposable bins clearly marked and suitably located.	---
165.	Are drip trays positioned wherever necessary?	Not evident	138. Drip trays to be provided wherever necessary.
166.	Do you have adequate localized extraction and scrubbing facilities for dust, fumes and gases? Please specify.	Scrubber installed. Detailed information available with HSE and concern plant.	---
167.	Whether walkways are clearly marked and free from obstruction?	Walkways were not marked.	139. It is recommended that walkway to be clearly marked and free from obstruction all the time.
168.	Do you have any inter-departmental competition for good housekeeping?	Regular cleaning done.	---
169.	Has your organization elaborated good	Housekeeping needs improvement.	140. Good housekeeping practices and

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	housekeeping practices and standards and made them known to the employees?		standards should be documented and to be made known to the employees.
170.	Are there any working conditions, which make the floors slippery? If so, what measures are taken to make them safe?	Working conditions, which make the floors slippery was available and adequate measures taken.	141. Efficient floor cleaning programs to be introduced.
171.	Does the company have adequate measures to suppress polluting dust arising out of materials stored on the roadside?	Some unwanted material found on road.	142. Internal roads to be kept clean & clear. Unwanted material to be removed from internal road.
	C-19.2 Machine and General Area Guarding (The GFR – Rule 54, The FA 1948 Section – 21, 22.)		
172.	Whether machinery and equipment which can cause physical injuries to operator have been identified?	Machinery & equipment which can cause physical injuries to operator was not identified.	143. Machinery & equipment which can cause physical injuries to operator to be identified.
173.	Are all moving parts and point of operation of machinery adequately guarded?	We observed that at some locations guard was missing.	144. All moving parts to be adequately guarded.
174.	Are all fixed guards securely bolted in position and in good condition?	Fixed guards were securely bolted in position and in good condition.	---
175.	Are all interlock guards for prevention of physical injury in good condition?	Interlock guards were not provided for prevention of physical injury.	145. Interlock guards to be provided wherever necessary.
176.	Are all emergency stop buttons effective and clearly labelled?	Emergencies stop buttons clearly labelled.	146. Ensure that all emergency stop buttons to be labelled properly.
177.	Are the operators for machines having moving parts aware of the danger of working with loose clothing?	Operators for machines having moving parts aware by training of the danger of working with loose clothing.	---
178.	Are the openings where there is free fall hazard covered or fenced securely?	No openings available where there was free fall hazard.	---
	C-19.3 Material Handling (The FA 1948 section 28,29,34, The GFR – Rule 58,59,60,62)		
179.	Are adequate equipment available for handling	Forklift, Stacker, hand trolley, etc. available for	147. Ensure that material handling equipment to

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	materials?	handling materials. Manual material handling also carried out.	be regularly inspected and maintained.
180.	Are the workers aware of the hazards associated with material being handled?	Workers found aware by training of the hazards associated with material being handled.	---
181.	Where manual handling is necessary, are the workers been trained? Do they practice this? Are workers warned for lifting of excessive weight? (Maximum weight of material for adult male and female are 55 Kg and 30 Kg respectively)	Workers trained where manual handling necessary. All workers warned for lifting of excessive weight.	---
182.	Do workers follow safe procedures for storage of materials?	Workers followed safe procedures for storage of materials.	---
183.	Is the register maintained to record particulars of examination of all lifting machines, tools and tackles?	Register was maintained to record particulars of examination of all lifting machines, tools and tackles. (Ref.: Form No. 10 available and valid)	---
184.	Are all the statutory examinations and tests carried out and certified by competent person(s)?	All the statutory examinations and tests were carried out and certified by competent person(s) by DISH Gujarat.	---
185.	Are the operators of crane, lifts, hoists and other mechanized operations adequately qualified?	We observed that operators of mechanized operations adequately qualified.	---
186.	Is the safe working load clearly marked?	SWL was not marked on the equipment	148. To be ensured Safe working load (SWL) clearly marked.
187.	Has the person employed to operate crane, forklift, or to give signals to crane been medically examined for eyesight and colour vision?	During periodic medical examination eye-sight and colour blindness was checked.	---
188.	Is the frequency of eyesight and colour vision examination as per the	Frequency of eyesight and colour vision examination as per the latest rules.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	latest rules?		
	C-19.4 Electrical Safeguarding		
189.	Are licensed electricians available for electrical work?	We came to know that Licensed electricians were available for electrical work.	---
190.	Whether area classification for electrical equipment has been carried out?	Flame proof electrical installation installed.	149. Hazardous Area Classification study to be carried out for electrical equipment's.
191.	Do the electrical fittings conform to area classification for electrical equipment?	We came to know that Electrical fittings were conform to area classification for electrical equipment.	150. Ensure that Electrical fittings shall conform to area classification for electrical equipment.
192.	Is a ground fault current interrupter system (ELCB) in use?	ELCB in use.	---
193.	Are all connections made by using appropriate plugs, receptacles or enclosures? Are fuses provided?	All connections made by using appropriate plugs.	---
194.	Are there any makeshift connection bare wires or damaged cables?	Shift connection bare wires or damaged cables was not observed.	---
195.	Is there a system of ensuring periodical inspection of hand tools, extension boards used for electrical work?	Visual inspection system followed before use of electrical equipment.	---
196.	Do the workers use proper types of PPE during the working on live line?	Workers used proper types of PPE during the working on live line.	151. Ensure that Electrical Safety Shoes, Safety Helmet, Safety Goggles, Arc Flash Suit of suitable rating 11KV hand glove and Shoes, Hot line stick tester used.
197.	Is the separate work permit issued for working on high voltage line?	No separate work permit issued for working on high voltage line.	152. Separate Permit to be issued for high voltage Electrical line. 153. Review CEA regulation.
198.	Whether the process(s) and equipment that generate and accumulate static charge have been identified?	Most of Processes, operations and equipment generating static charge were identified.	154. Process(s) and equipment that generate and accumulate static charge to be identified

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
199.	Whether all such equipment including pipelines for flammable materials are properly bonded and earthed?	Such equipment including pipelines for flammable materials were properly bonded and earthed.	155. Double earthing to be ensured.
200.	Whether earth pit resistance is measured, and the record maintained?	Earth pit resistance was measured and the record was maintained by electrical department.	---
201.	Whether lightning arrestor has been installed and is adequate?	Lightning arrestors were not installed.	156. Lightning arrestor survey to be carried out and accordingly lightning arrestors to be installed.
	C-19.5 Safety in Storage and Warehousing		
202.	Whether the Material Safety Data Sheet for all chemicals is available?	Material Safety Data Sheet for chemicals was available.	---
203.	Are the chemicals stored as per their hazardous properties including the incompatibility?	Most of the Chemicals stored as per their hazardous properties including the incompatibility.	157. Chemical compatibility chart to be displayed
204.	Are all containers clearly, indelibly labelled? Are all chemicals stored as per safety regulations?	The containers clearly, indelibly labelled & Identification was available on container.	158. Hazard symbol to be displayed on container.
205.	Whether all racks and steel cages have sufficient load bearing capacity?	The racks and steel cages have sufficient load bearing capacity.	159. SWL to be marked on racks.
206.	Is adequate natural ventilation provided to storeroom? Is there any emergency exit?	Adequate natural ventilation provided to storeroom. Emergency exit provided.	---
207.	Whether adequate firefighting arrangement existing in flammable chemical storage?	Fire Extinguisher, sand buckets, etc. available.	---
208.	Whether methodology for handling spillages of hazardous chemical available along with the equipment required handling the spillage?	Yes, sand available.	160. Spill kit to be provided for handling spillages of hazardous chemicals.
209.	Whether aisles are marked, and emergency exits displayed?	Aisles were marked and emergency exits were not displayed.	161. Emergency exits to be marked and displayed properly.
	C-19.6 Hazard Assessment for New Equipment		

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
210.	What is the system for effecting any change in the existing plant, equipment?	Informal system available for effecting any change in the existing plant, equipment was available.	162. MOC / Change Control Process to be established.
211.	Is there system for evaluating hazards from new equipment?	Formal procedure was not evident.	163. Formal system to be developed for evaluating hazards from new equipments.
212.	Whether the P and I diagrams, and other related documents are updated accordingly?	We came to know that P and I diagrams, and other related documents were updated accordingly.	---
213.	Is any Job Hazard Analysis (JHA) carried out after installation of new equipment?	Informal JHA carried out for non-routine activity.	164. JHA to be carried out after installation of new equipment.
	C-19.7 Hazards from Radiation Sources		
214.	Whether licenses have been obtained for storage /handling of radioactive material?	We came to know that storage of radiation sources was not available. As & when required (For NDT testing or other work) handling of radioactive material used by third party, they had valid license for handling of radioactive material.	---
215.	Whether approved Radiological Safety Officer appointed?	As & when required (For NDT testing or other work) approved Radiological Safety Officer appointed.	---
216.	Whether appropriate PPEs are used against radiation hazards?	As & when required (For NDT testing or other work) Radiological Safety Officer used appropriate PPEs against radiation hazards.	---
217.	Is the flooring of the radioactive material handling area amenable for proper decontamination?	Flooring of the radioactive material handling area was proper decontamination by third party which carried out job.	---
218.	Is the storage room of radiation source as per the licence condition?	They don't have storage of Radioactive material in the premises.	---
219.	Are all persons working in the facility have radiation safety training?	We came to know that as & when required NDT testing or other work only trained persons & qualified person	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
		handled working.	
220.	Is the operator handling devices using radioactive materials qualified and possess the necessary certificate?	As & when required (For NDT testing or other work) only trained persons & qualified person handling of radioactive materials.	---
221.	Is the periodical radiation monitoring carried out?	As & when required (For NDT testing or other work) radiation monitoring carried out	---
222.	Are the records of inventory of radioactive material maintained in the standard format and submitted to the competent authority as per the period specified?	We came to know that third party (which carried out job) was do all records of inventory of radioactive material maintained in the standard format and submitted to the competent authority as per the period specified	---
223.	Are emergency handling tools available?	We came to know that third party (which carried out job) had emergency handling tools available.	---
224.	Are the personnel monitoring badges (TLD, Pocket dosimeter etc.) assigned and worn by each radiation worker?	We came to know that third party (which carried out job) worn suitable personnel monitoring badges.	---
225.	Are the radiation symbol and red light displayed as required?	We came to know that, radiation testing done in lunchtime & all area cordon by barricaded line.	---
	C-20 Chemical Hazard		
	C-20.1 Transportation of Hazardous Substances		
226.	What potentially hazardous materials are transported to or from the site (including wastes)	List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Toluene, Sulphuric Acid, Xylene, Caustic lye, Methanol, Para formaldehyde, Formaldehyde-37%, Cumene, Phenol, Isopropyl Alcohol, Hydrogen Peroxide-50%, MCHA, Propyalzol Alcohol, Hydrochloric Acid, Acetophenone)	---
227.	What mode of transport are used?	Mode of transport was used	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	1) Road, 2) Rail, and 3) Pipelines	• Road • Pipeline	
	1) Road		
228.	Does the company employ licensed vehicle of its own / outside sources?	Outside sources vehicle.	---
229.	Are the loading / unloading procedures in place and safety precautions displayed?	Loading / unloading procedures in place but safety precautions were not displayed.	165. Loading / unloading procedures to be displayed at conspicuous places.
230.	Is there a provision to check the healthiness of road tanker with respect to explosives rules?	Checklist available for check the healthiness of road tanker with respect to explosives rules.	---
231.	Are loaded tankers or trucks parked in a specific area on-site?	Loaded tankers or trucks parked in a specific area on-site.	---
232.	Do all truck and tanker drivers carry transport emergency (TREM) card or instruction booklet?	Truck and tanker drivers carried out transport emergency (TREM) card or instruction booklet.	---
233.	Do all truck and tanker drivers get training in handling emergencies during transport?	Tanker driver carrying hazardous materials were trained by approved training institute.	166. Ensure that all truck and tanker drivers trained in handling emergencies during transport (As per rule 9 of MV Act).
234.	Are all the tankers marked for proper Hazchem code?	Tankers marked for proper Hazchem code.	---
	2) Rail		
235.	What hazardous materials are transported by rail?	No Hazardous materials transported by rail.	---
236.	Does the company have a direct siding on site?	Not Applicable	---
237.	Are tankers or other wagons used in transportation?	Not Applicable	---
	3) Pipelines		
238.	What materials are transported to and from the site by pipelines?	Natural Gas pipelines	---
239.	Are the pipelines underground or over ground?	Pipelines were above ground	---
240.	Are corrosion protection	Yes, painting done	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	measures employed in pipelines?		
241.	Whether intermediate booster pumps are used?	No	---
242.	What is the maximum, minimum and average transfer rates?	Details available with concern department	---
243.	Are the pipelines extended in the public domain?	Pipelines were not extended in the public domain.	---
244.	Are the pipelines dedicated for each type of chemicals?	Yes, pipelines dedicated for each type of chemicals	---
245.	Are the pipelines fitted with safety equipment such as leak detectors, automatic shut-off valves etc.?	Manual shut-off valve was available.	---
246.	What is the frequency and method of testing of the pipeline?	Records available with maintenance Department.	---
247.	Is there written procedure for tackling leakages in pipeline?	Not evident	---
	C-20.2 Handling of Hazardous Substances		
248.	What are the hazardous substances handled in the factory?	List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Toluene, Sulphuric Acid, Xylene, Caustic lye, Methanol, Para formaldehyde, Formaldehyde-37%, Cumene, Phenol, Isopropyl Alcohol, Hydrogen Peroxide-50%, MCHA, Propyalzol Alcohol, Hydrochloric Acid, Acetophenone)	---
249.	Whether quantity of hazardous substances is above the threshold limit specified in the Manufacture, Storage and Handling of Hazardous Substances Rule, 1989? If yes, then required documentation is available as per the rule.	Quantity of hazardous substance was not above the threshold limit specified in The Manufacture, Storage and Handling of Hazardous Substances Rules, 1989.	---
250.	Whether written procedure for handling the hazardous substance is available and	Work Instructions were prepared for safe handling of chemicals / substances.	167. Periodic safety training on safe handling of chemicals

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	operators are trained for handling such substances including actions required in case of leakages and spillages?		should be imparted to workers by DISH approved training institute.
251.	Are the employees aware of the hazards arising from hazardous substances and safety precautions to be taken during handling of these?	Most of Employees aware of the hazards arising from hazardous substances and safety precautions taken during handling of hazardous substances.	---
	C-20.3 Material Safety Data Sheets (MSDS)		
252.	Are the material safety data sheets available for all the chemicals handled, used and manufactured in the factory?	Material safety data sheets (MSDS) available for the chemicals handled, used and manufactured in the factory.	---
253.	Whether the latest MSDS are displayed at strategic locations?	Some place MSDS were not displayed.	168. Abstract of MSDS should be prepared in the language understood by majority of workers and display at conspicuous places.
254.	Is it available in local language?	MSDS abstract available in English only.	169. To be made available in local language.
	C-20.4 Spill Control Measures		
255.	Whether spill control procedure is available?	Dry sand provided and collection pit available.	170. Spill control procedure to be prepared and displayed.
256.	Whether spill collection pit / sump is available at the workplace?	Spill collection pit / sump was available at the workplace.	---
257.	Whether methodology for recovery / disposal of collected material has been established?	Methodology for recovery / disposal of collected material was established.	---
	C-20.5 Storage of Hazardous Substances		
258.	Whether storage vessels are identified with the capacity as required under MSIHC, Rules 1989.	List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Toluene, Sulphuric Acid, Xylene, Caustic lye, Methanol, Para formaldehyde, Formaldehyde-37%, Cumene, Phenol, Isopropyl Alcohol, Hydrogen Peroxide-	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
		50%, MCHA, Propyalzol Alcohol, Hydrochloric Acid, Acetophenone)	
259.	What are the storage pressure and temperature?	Kindly see para 6.2 "List of Hazardous Raw Materials". For storage pressure and temperature.	---
260.	Whether vessels are above ground / underground?	Vessels were above ground as well as underground.	---
261.	If any of the tanks storing flammable material, whether electrical equipment and fittings are as per electrical area classification?	We came to know that electrical fittings are flameproof.	171. Ensure that electrical fittings are as per Hazardous area classification.
262.	Is the bunded area takes into account the total quantity of the largest tank?	Adequate bunded area.	---
263.	Whether the bund perimeter takes into consideration of trajectory of leak from tank?	Trajectory of leak from tank was considered most of storage tanks.	---
264.	Are the vessels properly bonded and earthed and whether periodically checked and record maintained?	The vessels properly bonded and earthed and regularly checked.	172. Double earthing to be provided.
265.	Are the vessels fitted with remotely controlled isolation valves?	Vessels were not fitted with remotely controlled isolation valves.	---
266.	Are vessels provided with emergency vent, relief valve, bursting disc, level indicator, pressure gauge, overflow line?	As applicable, installed. Data available with concerned plant / department.	---
267.	Where do such vents discharge?	Scrubbers available	---
268.	Are the vessels provided with alarms for high level, high temperature and high pressure?	Vessels were not provided with alarm for high level, high temperature and high pressure.	173. Alarm to be provided for high level, high temperature & high pressure to vessels.
269.	Are standby empty tanks or any other alternate systems provided for emptying / transfer in case of emergencies?	Standby empty tank was not evident.	174. Standby empty tank to be provided.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
270.	What are the provisions made for firefighting / tackling emergency situations around the storage vessels?	Following provisions made for firefighting / tackling emergency situations. • Fire hydrant System - Partial • Fire Extinguishers • Sand Buckets	---
271.	Has any consequence analysis for loss of containment been carried out?	Consequence analysis for loss of containment was not carried out.	175. Consequence analysis for loss of containment to be carried out.
272.	Whether the vessels are tested as per statute?	The vessels were tested as per statute.	---
273.	Whether log sheets are filled up on daily basis for recording the parameters of these vessels?	Log sheets were filled up on daily basis for recording the parameters of these vessels.	---
274.	Whether monitors for detection of leakage of flammable / toxic material installed?	Portable LEL detector, O_2 detector, Multi gas detector, etc. available	---
275.	Whether the chemicals stored are as per their compatibility?	Chemicals stored as per their chemical compatibility chart.	---
	C-20.6 Gas Cylinders		
276.	What are the various gas cylinders used in the plant?	Nitrogen, H_2, zero air, O_2, DA, etc.	---
277.	Are valid licenses available for storing all these cylinders?	License not required.	---
278.	Are the cylinders stored and segregated as per their compatibility?	The cylinders stored and segregated as per their compatibility.	---
279.	What are the measures taken for combating any emergency in the cylinder's storage area?	Fire extinguishers were available for combating any emergency in the cylinder's storage area.	---
280.	Whether integrity test certificates are obtained from the suppliers of the cylinders?	Its informed that integrity test certificates were obtained from the suppliers of the cylinders.	---
281.	Are the cylinders chained and secured properly along with the valve caps and proper identification colour code?	Most of cylinders chained and secured properly along with the valve caps.	---
282.	Are the cylinders protected	Shed provided for the	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	from heat or sun and rain?	cylinder protection from heat or sun and rain.	
283.	Whether monitors for detection of leakage of flammable / toxic gas installed?	Portable LEL detector, O_2 Detector, Multi gas detector, etc. available	---
	C-20.7 Labelling and Colour Coding		
284.	Are all the containers, vessels and storage tanks labelled for its content and capacity?	The containers, vessels and storage tanks labelled for its content and capacity.	176. To ensure that all the containers, vessels and storage tanks labelled for its content and capacity.
285.	Whether the pipelines are colour coded as per IS 2379?	The pipelines were colour coded as per IS 2379.	177. Ensure that pipelines are colour coded as per IS 2379.
286.	Is any plant specific colour code followed?	Plant specific colour code was followed.	---
287.	Whether the colour codes are displayed conspicuously in the working areas?	The colour codes were not displayed conspicuously in the working areas.	178. Colour code to be displayed conspicuously in the working area.
	C-20.8 Hazardous Waste Management		
288.	Is identification done for various types of hazardous wastes?	Identification done for various types of hazardous wastes as per GPCB Norms.	---
289.	Are these quantities less than those specified by the Hazardous Wastes (Management & Handling) Rules, 1989?	We came to know that quantities less than those specified by the Hazardous Wastes (Management & Handling) Rules, 1989.	---
290.	What are their disposal modes?	Their disposal modes as per GPCB Norms. Waste disposal permission was evident.	---
291.	What are the systems / measures adopted for controlling air / water / land pollution?	The systems / measures adopted for controlling air / water / land pollution as per GPCB Norms.	---
292.	Whether the solid waste like combustibles, plastic, metals etc. segregated?	The solid waste like combustibles, plastic, metals etc. were segregated.	---
	C-21 Fire and Explosion Hazard		
	C-21.1 Organizational Set-up for Fire Fighting		
293.	What is the total strength of fire station and fire crew?	Mutual Aid with Nearest Fire Station available.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
294.	How many fire crews are available in each shift?	No dedicated fire crew in the factory.	---
295.	Is there fire squad identified in each shift?	Fire-fighting trained persons were available in each shift.	---
296.	Standing fire order is available with latest revision	Standing Fire order was not available.	179. Standing Fire order to be obtained from NBC 2016 and display at conspicuous locations in plant and other areas.
297.	How is the communication with fire station?	Communication with external fire brigade by telephone / mobile.	---
298.	Does fire safety inspections carried out?	Fire safety inspections carried out informally.	---
299.	Does emergency procedure available for leakage or combustion of flammables?	Dry sand available.	---
300.	What measures are available to control the fire load in the plant area?	Fire extinguisher available to control the fire load in the plant area.	---
301.	Whether technical knowledge and skills of the manager and staff responsible for overall fire safety of the plant is adequate?	Qualified and experienced managers and staff was deployed for operation of the process plant.	---
302.	How many major and minor incidents / fires were there in the factory during the last five years? Give department / plant wise.	Data available at concern department.	---
303.	Have all the fires / incidents been investigated and corrective actions taken? Give break-up.	Fires / incidents been investigated and corrective actions taken. Data was available at EHS department.	---
304.	**Resources:** 1. Adequacy of protective clothing (coat, trouser, gloves, boots and helmets); 2. Availability of SCBA for firefighting operations and spare cylinders (at least 2 for	1. Adequate 2. SCBA set not available 3. Adequate 4. Adequate	180. SCBA set with spare cylinders (at least 2 for each SCBA Set) to be provided.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	each SCBA); 3. Adequacy of hose, nozzles, ladders, lighting equipment and pumps; and 4. Communication facility at fire station, walkie talkie sets during firefighting.		
	C-21.2 Built in Safety in Civil Design and Construction		
305.	Whether the two-safe means of escape available? Are they in separate directions?	Two safe means of escape available in separate directions.	---
306.	Is emergency exits provided to the building handling flammables?	Emergency exits provided to the building handling flammables.	---
307.	Whether emergency lights are provided?	Emergency lights were not provided.	181. Emergency light to be provided.
308.	Whether fire / smoke detectors are installed in fire prone areas?	Smoke detectors were not installed in fire prone areas.	182. Smoke detectors to be installed in fire prone areas.
309.	Whether fire call points are provided in different areas?	Fire call points were not seen.	---
310.	Whether Fire hydrants are provided near the buildings?	Fire hydrants points were available.	---
311.	Is ventilation system in plant handling flammables is adequate to prevent formation of flammable mixtures?	Ventilation was adequate to prevent formation of flammable mixtures.	---
312.	Is adequate separation is provided between combustible / flammable materials and other material to restrict the fire growth?	Adequate separation was provided between combustible / flammable materials and other material to restrict the fire growth.	---
313.	Access routes for firefighting operations is available for areas having high fire load	Access routes for firefighting operations was available for areas having high fire load.	---
314.	Whether building changes interferes with fire detection and / or fire suppression systems?	No changes made in building after installation of Fire Suppression System.	---
315.	Whether building changes cause unreasonable fire	No fire rated walls available.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	loading / openings in the fire rated walls?		
	C-21.3 Built in Safety in Electric Circuits and Equipment		
316.	Are the electrical equipment in areas where flammables mixture is likely to be present of flame-proof type?	Flameproof type electrical installations were provided.	183. Ensure that Electrical equipment and fittings to be as per electrical area classification.
317.	Are lightning arrestors are provided to the buildings / structures storing flammable materials?	Lightning arrestors were not installed.	184. Adequate lightning arrestor to be provided to the buildings / structure storing flammable materials.
318.	Whether adequate bonding and grounding of electrical equipment / pipelines provided?	Bonding and grounding were provided in electrical equipment / pipelines.	---
	C-21.4 Explosive Substances (Explosive Act ,1884 and the Explosive Rules,1983)		
319.	Whether necessary license / approval taken from concerned statutory bodies?	No explosive substance used in factory.	---
320.	Whether systems for explosion suppression, high speed fire detection with deluge, sprinklers, explosion venting etc. are provided?	Not Applicable	---
321.	Whether explosion resistant walls or barricades are provided around explosive storage?	Not Applicable	---
322.	Whether explosive substance storage areas are restricted for entry?	Not Applicable	---
323.	Whether only trained persons are handling explosive substances?	Not Applicable	---
324.	Whether explosive substances are stored and transported in approved containers only?	Not Applicable	---
325.	Whether electrical fixtures in areas handling explosives are explosion proof type?	Not Applicable	---
326.	Whether adequate measures are taken to	Not Applicable	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	prevent any sources of ignition where explosive substances are handled?		
	C-21.5 Fire Safety in Handling Flammable and Explosive materials		
327.	Whether emergency procedure is available for control of leakage?	Emergency procedure was available in OSEP. Dry sand was available.	---
328.	Whether emergency measures are displayed locally in case of accidental spillage / leakage?	Precautionary measures displayed.	185. Emergency measures to be displayed in case of accidental spillage / leakage.
329.	Whether facility is provided for safe drainage of combustible or flammable liquids in case of leakages?	Dyke provided for safe drainage of chemicals in case of leakages.	---
330.	Whether highly flammable liquids are stored under inert atmosphere?	No material stored under inert atmosphere.	---
331.	Whether flammable storage tanks are provided with flame arrestors?	Flammable storage tanks were provided with flame arrestors.	---
332.	Whether suitable PPEs are provided?	Suitable PPEs were provided	---
	C-21.6 Fire Detection and Alarm System		
333.	What type of fire detection and alarm system provided?	Fire detection & alarm system were not installed.	186. Suitable fire detection & alarm system to be installed.
334.	Whether all fire prone areas of the plant are covered with fire detection system?	Fire detection & alarm system were not installed.	187. Fire prone areas of the plant to be covered with fire detection system.
335.	Whether fire detection equipment and smoke alarms in good operating condition?	Fire detection & alarm system were not installed.	---
336.	Whether the number of fire call points are adequate and free from obstruction?	Fire call point was not seen.	---
337.	Whether regular inspection / maintenance / testing of fire detection and alarm system carried out and records maintained	Fire detection & alarm system were not installed.	---
338.	Whether any atmospheric monitoring is carried out for explosive mixture of gases or vapours?	Workplace monitoring reports was not evident.	188. Workplace monitoring to be carried out at regular intervals and

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
			records to be maintained.
339.	Whether emergency power supplies are provided to fire detection and fire alarm system?	DG set available for emergency power supplies.	---
340.	Whether smoke detectors are located considering ventilation pattern?	Fire detection & alarm system were not installed.	---
341.	Whether annunciation of fire is local or in the control room or in both places?	Fire detection & alarm system were not installed.	---
342.	Whether fire panel is constantly attended?	Fire detection & alarm system were not installed.	---
	C-21.7 Passive and Active Fire Protection System		
343.	What are the passive fire protection measures available? (Barriers, doors, dampers etc.)	Passive fire protection was not observed.	189. Passive fire protection measures to be provided wherever required.
344.	Are the areas requiring fire barriers identified?	Area was not identified.	---
345.	Whether the fire barrier provided is of adequate ratings?	Fire barrier was not provided.	190. Fire barrier to be provided.
346.	Whether ventilation ducts in flammable areas have been provided with isolation dampers of suitable fire rating?	Ventilation ducts in flammable area was not provided with isolation dampers.	191. Isolation dampers to be provided in ventilation ducts.
347.	Whether sprinklers / deluge are installed wherever necessary?	Sprinkler system was not seen.	192. Review Rule 66A of the GFR-1963 for requirement of Sprinkler.
348.	Whether regular inspection / maintenance / testing of fire protection system carried out and records maintained?	Regular inspection of fire protection system was carried out and records were maintained.	---
	C-21.8 Fixed Fire Extinguishing System		
349.	What are the sources of firewater and whether they are dedicated to the fire extinguishing system?	Common storage tank provided for fire, domestic and process use with supply line from GIDC.	193. Dedicated fire water storage to be provided.
350.	Whether the capacity of dedicated water reservoir is adequate to supply to hydrants for minimum 2 h?	Its informed that the capacity of dedicated water tank was adequate to supply to hydrant for	194. Recommended to keep adequate storage of water for exclusively for fire.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
		minimum 2 hrs.	
351.	Whether un-interrupted power supply is provided to the firewater pumps?	DG set available	---
352.	Whether the extinguishing medium selected is appropriate to the class of fire (water, gaseous, foam, dry powder)?	The extinguishing medium selected was appropriate to the class of fire.	---
353.	Whether fire hydrants layout is available?	Yes, Fire hydrants layout was not evident.	195. Fire hydrant layout to be made available.
354.	Whether additional (over minimum requirement) fire hoses, nozzles are available?	Additional (over minimum requirement) fire hoses, nozzles were not available.	196. Additional (over minimum requirement) fire hoses, nozzles to be provided.
355.	Whether the hydrants lines are kept pressurized?	No	197. Fire hydrant lines to be kept pressurized.
356.	Whether regular inspection / maintenance / testing of fixed fire extinguishing systems carried out and records maintained?	Informal inspection of fixed fire extinguishing systems was carried out by internally.	---
	C-21.9 Portable Fire Extinguishing System		
357.	Whether suitable type and numbers of fire extinguishers provided?	Total 53 Nos. of Fire Extinguishers installed in entire premises.	---
358.	Whether the fire extinguishers are located at conspicuous position and easily accessible? Are they fully charged and tagged?	The fire extinguishers were located at conspicuous position and easily accessible & they fully charged and tagged.	---
359.	Whether fire extinguishers periodically inspected, tested, refilled and records maintained?	Fire extinguishers were periodically inspected, tested and refilled and its records was also available.	---
360.	Whether defective / unchecked fire extinguishers present at site?	Defective fire extinguishers were not present at site.	---
361.	Whether additional fire extinguishers are available?	Additional fire extinguishers were available.	198. Ensure that additional 10% fire extinguishers are available.
	C-21.10 Fire Fighting Equipment and Facilities		
362.	Whether fire tenders (water / foam) are available?	Fire tender was not available.	---
363.	Whether the fire-fighting	The fire-fighting system	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	system and equipment approved, tested and maintained as per relevant standard?	and equipment approved, tested and maintained as per relevant standard.	
364.	Whether the SCBA / fire suit provided to firefighting team for fire action?	Not evident	199. SCBA set with spare cylinders (at least 2 for each SCBA Set) to be provided.
365.	What is system for maintenance/ recharge of SCBA?	SCBA set was not evident.	---
366.	Is proper access available for fire-fighting equipment?	Proper access available for fire-fighting equipment.	200. Make sure that access are always unobstructed.
367.	Whether fire hose cabinets are in good condition, easily visible, and accessible?	Fire hose cabinets were in good condition, easily visible and accessible.	---
368.	Whether drill tower is available? Are fire personnel carrying out regular fire drill?	Drill tower was not available. Firefighting training was carried out regular interval.	---
369.	What is the communication facility at fire station? Is it adequate?	Communication facility at fire station: Telephone / mobile	---
	C-21.11 Fire Drill		
370.	Whether mock fire drills are conducted? What is the frequency of drills?	Mock Drills conducted. Last drill – 12/08/2023	201. Firefighting drill to be carried out at every 2 months as per rule 66-A of the GFR-1963.
371.	Whether fire drills are also performed in night Shift?	Fire drill was not performed in night Shift.	202. Fire drill to be performed in night shift.
372.	Whether feedback of fire drill is documented?	Mock drill feedback records was documented.	---
373.	What is the system of mutual-aid scheme?	Mutual aid agreements with nearby fire stations & industries	---
	C-21.12 Fire Fighting Training		
374.	Whether there is a system of providing fire- fighting training to plant personnel?	Yes, system of providing firefighting training to plant personnel was available.	---
375.	What is the frequency and duration of such training? Whether training records are maintained?	At regular interval, firefighting training was provided.	203. Firefighting training records to be maintained.
376.	Whether fire squads are	Fire squad was not	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	identified for different areas for first-aid firefighting and rescue, and suitably trained?	identified.	
377.	Are all personnel conversant with the fire prevention and protection measures?	Most of personnel were familiar with the fire prevention and protection measures to be taken.	---
378.	Whether the fire staff are sent for refresher / advanced training courses?	Firefighting staff was not sent to refresher / advanced training course.	204. Fire staff to be sent for refresher / advanced training courses.
	C-21.13 Static Electricity and Lightning		
379.	Whether all vessels and pipes are provided with suitable bonding and grounding?	Most of the vessels and pipes were provided with suitable bonding and grounding.	205. Double earthing to be ensured.
380.	Whether arrangement has been made for grounding the tanker containing flammable liquid during loading / unloading?	Earthing available for grounding the tanker containing flammable liquid during loading / unloading.	---
381.	Whether spark resistant tools are provided?	Spark resistant tools were not provided.	206. Spark resistant tools to be provided and ensure that Spark resistant tools used every time.
382.	Whether lightning protection is provided and is adequate?	Lightning Arrestors installed.	207. Make sure that installed lightning arrestors are adequate.
383.	Whether antistatic clothing, hand gloves and footwear are provided?	Antistatic clothing, hand gloves and footwear were not provided.	208. Antistatic clothing, hand gloves and footwear to be provided as and when required.
	C-21.14 Pressure Relief System (The FA 1948 section 31, The GFR – Rule 61)		
384.	Whether the listing of all 'pressure plants' [as defined under Factories Act] has been done?	Pressure plant area identified and pressure plant list maintained by concern department.	---
	C-22 Industrial Hygiene / Occupational Health		
	C-22.1 Ventilation, Illumination, Noise, Vibration, Heat stress and Non-ionizing Radiations		
	C-22.1.1 Ventilation		
385.	Whether any ventilation study has been carried out?	Ventilation study was not evident. Natural ventilation	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
		was adequate in plant.	
386.	Whether natural ventilation is adequate or not?	It was informed that natural ventilation was adequate in plant.	---
387.	Whether dust / fumes / hot air is generated in the process?	Dust / Fumes generated in the process.	---
388.	Is there any exhaust ventilation system in any section of the plant?	Natural ventilation system in plant.	---
389.	Is periodic / preventive maintenance of ventilation system carried out and record is maintained?	As natural ventilation so, no such records was evident.	---
390.	Does any ventilation system re-circulate the exhausted air in work areas?	Natural ventilation system in plant.	---
391.	Is the work environment assessed and monitored for chemical and physical hazards?	Not evident	209. Workplace monitoring should be done, and record should be maintained in Form-37.
392.	Whether PPE are provided to workers exposed to dust / fumes and gases?	Adequate PPEs provided to workers	---
	C-22.1.2 Illumination		
393.	Whether illumination study has been carried out for the assessment of illumination level?	Illumination study was not evident.	210. Illumination study to be carried out and records to be maintained.
394.	Is there any system of periodical cleaning and replacing the light fittings / lamps in order to ensure that they give the intended illumination levels?	System available for periodical cleaning and replacing the light fittings / lamps in order to ensure that they give the intended illumination levels.	---
395.	Are the workers subject to periodic optometry tests and records maintained?	Periodic optometry test was carried out for the workers during periodical medical check-ups.	---
396.	Are emergency lighting available at first aid centre?	No first aid centre.	---
	C-22.1.3 Noise (The GFR 102 Schedule 23)		
397.	Whether any noise study conducted?	Noise study conducted by third party & records maintained.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
398.	Are there any machines / processes generating high noise?	High noise area i.e., Compressor, Blower, etc.	---
399.	Whether engineering and administrative controls been implemented to reduce noise exposure below the permissible limits?	Engineering and Administrative Control: Noise monitoring, PPE's, training, etc.	---
400.	Is there a system of subjecting all those employees to periodic audiometric test who work in high- level noise areas?	Periodic Audiometric test was carried out for high level noise area.	---
401.	Whether the workers are made aware of the ill effects of high noise?	Workers were aware of ill effects of high noise.	---
402.	Whether earmuffs / plugs are provided and used?	Earmuffs / plugs were provided and used.	---
	C-22.1.4 Vibration (The GFR 102 schedule 23)		
403.	Are there equipment which contributes excess level of vibrations and whether they are identified?	No such equipment identified.	211. Identify vibrating equipment and vibration study to be carried out wherever necessary.
404.	Whether any vibration study has been carried out?	Vibrations study was not evident. During maintenance, vibration measures informally.	---
405.	Are the measures taken to combat vibration to acceptable levels?	Necessary measures taken to combat vibration to acceptable levels.	---
406.	What is the frequency for measurements of vibration?	During preventive maintenance, vibration measures informally.	---
407.	Are the records of measurements and maintenance of equipment / system maintained?	Records of maintenance of equipment / system maintained at concern department.	---
	C-22.1.5 Heat Stress / Cold stress (Extremes of Temperature)		
408.	Are there sources from equipment increasing the heat load in workplaces?	No source from equipment increasing the heat load in work places.	---
409.	Whether evaluation of heat stress is carried out?	Evaluation of Heat Stress was not done.	---
410.	Whether natural ventilation is adequate to minimize the	Natural ventilation was adequate to minimize the	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	heat stress in work environment?	heat stress in work environment.	
411.	Are resources available to deal with very hot or very cold conditions (drinking water, lined gloves, insulated boots)?	Appropriate PPEs, cold drinking water were provided. Proper engineering control & Administrative control made available.	---
412.	Do workers know the symptoms of heat cramps / heatstroke or frost bite / hypothermia?	Workers know the symptoms of heat cramps / heatstroke.	---
413.	Are the personal protective equipment suitable for reducing the effects of heat stress available?	The personal protective equipment suitable for reducing the effects of heat stress available	---
	C-22.1.6 Non-ionizing Radiations		
414.	Does the work involve likely exposure to non-ionising radiations (ultraviolet, infrared, radiofrequency, microwaves, lasers, etc.)	Non-ionizing radiations source was not available.	---
415.	Whether risk assessment have been done for all work areas involving presence of non-ionising radiations?	Not Applicable	---
416.	Are the work areas displayed with relevant safety signs?	Not Applicable	---
417.	Are the employees aware about the hazards of non-ionising radiations?	Not Applicable	---
418.	Do written procedures exists for working in non-ionising radiations?	Not Applicable	---
419.	Is the work environment monitored periodically for physical hazards and control measures initiated whenever deviation from permissible values is observed?	Not Applicable	---
420.	Whether suitable personal protective equipment are provided to workers exposed to non-ionising radiations?	Not Applicable	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	C-22.2 Workplace Monitoring for Hazardous Chemicals (Sec. 7-A FA 1948 & Rule 12-B of GFR, 1963)		
421.	Whether the dust, fumes, smoke, aerosols and mist are monitored as per statute and records maintained?	Workplace monitoring was not carried out.	212. Work place monitoring to be carried out and records to be maintained in form-37.
422.	What are the types of detectors used for monitoring concentration of hazardous chemicals?	Portable LEL detector, O_2 detector, Multi gas detector, etc. available	---
423.	Is any alarm system installed for any leakage of hazardous chemicals?	No alarm system provided for any leakage of hazardous chemicals	213. Alarm system to be installed for leakage of hazardous chemicals.
424.	Are antidotes available for toxic chemicals?	No antidotes available for toxic chemicals.	---
425.	Are control measures initiated whenever deviation from permissible values is observed?	Adequate control measures provided whenever deviation from permissible values is observed.	---
	C-22.3 First Aid Facilities and Occupational Health Centre (OHC) (The FA Sec.-45, Gujarat Factories Rules, 1963-Rule 68-U, 68-V, 71)		
426.	Are adequate numbers of first aid boxes provided? Give location details?	First Aid Box: 2 Nos. List available at EHS / HR Dept.	---
427.	Are qualified / trained first aiders available in each shift?	Qualified / trained first aiders were not available.	214. First aid training to be provided to plant personal for emergencies.
428.	How many qualified / trained first aiders are available at each plant / department?	Trained first aiders were not available.	215. Trained first aiders to be made available in each shift.
429.	How many persons are trained / given refreshers training in first aid in a year?	First aid training was not done.	---
430.	Whether occupational health centre is provided?	Occupational health centre was not available. Mutual aid with nearby hospital.	216. Kindly review rule 68-U of GFR-1963 for OHC.
431.	Does OHC conform to the provisions of the existing statutes?	Company maintained records of medical examination.	---
432.	Are the Medical Attendants / Doctors available in each shift?	No	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
433.	What facilities are available for transportation of the injured to hospital?	Emergency vehicle was available round the clock basis.	---
434.	Are the names of the trained first aiders displayed?	Trained first aider was not available.	217. First Aid training to be provided from competent party.
435.	Are the name of nearest hospitals and its telephone number available in OHC?	Name of nearest hospitals and its telephone number was available.	---
436.	Does the plant have any special preventive medicine program?	Plant have not conducted special preventive medicine program	---
437.	Is ambulance posted in proper place and is it available whenever required?	Emergency vehicle available round the clock	---
438.	Are sufficient numbers of anti-dotes available in case of any emergency?	Antidotes were not available for toxic chemicals. First aid box provided.	---
439.	Are fire safety measures provided in first aid centre?	No First Aid centre.	---
440.	Are emergency lighting arrangements available at first aid centre?	No First Aid centre.	---
	C-22.4 Periodic Medical Examination (The GFR Rule 68-T, Rule 102)		
441.	Whether the periodical medical examination of employees, required under relevant statute is carried out?	The periodical medical examination (Form No. 32) of employees, required under relevant statute was carried out.	---
442.	Whether it is ensured that contractors employee are medically examined during pre-employment as well as during the course of employment?	It was informed that contractors employee were medically examined during pre-employment as well as during the course of employment.	---
443.	During the periodical medical examination of the workers, are they examined as per the hazardous process in which they work? (First schedule of The Factories Act, 1948)	During the periodical medical examination of the workers, they were examined as per the hazardous process in which they work. (First schedule of The Factories Act, 1948)	---
444.	Are the records of all such examination maintained?	Records of all such examination was maintained.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	C-22.5 Personal Protective Equipment and Emergency Equipment		
445.	Whether list of required PPE for each hazardous activity is available?	Activity-wise PPE matrix was prepared.	---
446.	Whether feedback from workers obtained during selection of PPE?	Informal feedbacks were taken from workers during selection of PPE.	---
447.	Have the workers been trained in proper use of PPE including BA sets?	The workers been trained in use of PPEs.	218. Refresher training to be provided to all workers.
448.	What is the system of procurement, inspection, issue, maintenance and replacement of PPE?	System was available for procurement, inspection, issue, maintenance and replacement of PPE.	---
449.	Whether qualitative and quantitative fit-check for respirators is ensured prior to use?	PPEs checking was carried out before use.	---
450.	What are the arrangements for safe custody and storage of PPE?	Lockers provided for safe custody and storage of PPE.	---
451.	Are the contractor's workers provided with the required PPE?	Contractor's workers provided with the required PPE.	---
452.	Do the PPE conform to any standard?	PPEs conform to IS standard.	---
453.	Are sufficient eye wash fountains and safety showers available?	Sufficient eye wash fountains and safety showers available.	---
454.	Whether appropriate respiratory protective devices are available in accordance to the hazard potential?	Appropriate respiratory protective devices were available in accordance to the hazard potential.	---
455.	Are the staff members trained in the right uses of respiratory protective devices?	Staff members trained in the right uses of respiratory protective devices.	---
	C-22.6 Occupational Diseases		
456.	Whether pre-employment medical check-up data available?	Pre-employment medical check-up data not evident.	219. Pre-employment medical examination data to be made available in form-33.
457.	During the medical check-up, is any person found having occupational diseases mentioned in 3rd	During the medical check-up, any person not found having occupational diseases mentioned in 3rd	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	schedule of The Factories Act, 1948?	schedule of The Factories Act, 1948.	
458.	Whether the medical practitioner informed the Chief Inspector of Factories about the occurrence of the occupational disease?	No abnormality detected during periodic medical examination of workers.	---
	C-23 Accident / Incident Reporting, Investigation and Analysis (The GFR – Rule 103,111)		
	C-23.1 Accident Reporting and Database Management		
459.	What is the procedure for accident / incident /dangerous occurrence reporting?	Accident / incident / dangerous occurrence reporting register was available.	---
460.	Whether the accident data for the last five years for reportable and non-reportable accidents are available?	Accident data for the reportable and non-reportable were available. (Form 29 – Maintained)	---
	C-23.2 Accident Investigation		
461.	Are all the accidents investigated?	We came to know that the accidents were investigated.	---
462.	Whether accident investigation procedure is documented?	Informal procedure available.	---
463.	Whether accident investigation reports are submitted to top management?	Accident investigation reports were submitted to top management.	---
464.	How are the findings from accident investigation reports communicated to workers?	Learning from incident, training.	---
	C-23.3 Analysis of Accidents		
465.	Whether accident analysis is done as per IS 3786?	Accident analysis was done informally.	220. Accident analysis to be done as per IS 3786.
466.	Whether root causes of accidents are analysed?	Root causes of accidents were analysed informally.	221. Root cause of accident to be analysed.
467.	Is the accident statistics effectively utilized? If yes, how?	The accident statistics was not effectively utilized.	---
468.	What nature of injuries occurred during the last five years?	Data available with EHS department.	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	C-23.4 Implementation of Recommendations		
469.	How does the management ensure implementation of the recommendations to avoid recurrence of accidents and incidents?	Recommendation discussed in the meetings.	---
	C-23.5 Reporting and Investigation of Near miss Incidents		
470.	Are all near-miss incidents reported and investigated?	SOP for Near-miss incident reporting, investigation, analysis was not evident.	222. SOP for Near-miss incident reporting, investigation, analysis to be prepared. 223. Impart training to all workers to observed near miss on-site.
471.	Is there any system of classifying and analysing the near-miss incidents?	System of classifying and analysing the near-miss incidents was not available.	224. System to be developed to classifying & analysing the near-miss incidents.
	C-24 EMERGENCY PREPAREDNESS		
	C-24.1 Site Specific Details		
472.	Are the site area maps (including layout, access roads and assembly points) available in control room / emergency control centre?	The site area maps (including layout, access roads and assembly points) was available.	---
	C-24.2 Duties and Responsibilities of Key Personnel		
473.	Is the hierarchy of emergency response personnel right from site emergency controller downward, and alternative officials identified?	The hierarchy of emergency response personnel right from site emergency controller downward, and alternative officials was identified. (Ref.: - On-site emergency plan)	---
474.	Are the duties and responsibilities assigned to the designated officials during emergency, both during and outside normal working hours clearly identified and understood by them?	Duties and responsibilities assigned to the designated officials during emergency, both during and outside normal working hours clearly identified and understood by them. (Ref.: On-site emergency plan)	225. Ensure all employees including contractor workers are trained on On-site emergency plan
	C-24.3 Identification of Emergencies and Accident Scenario		
475.	Are the possible accident scenarios leading to emergency identified and	All possible accident scenarios leading to emergency identified and	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	known to the operating personnel?	known to the operating personnel. (Ref.: - On-site emergency plan)	
476.	Are approved emergency preparedness plans (on-site and off-site) in place?	Approved emergency preparedness plans (on-site emergency plan) was available.	---
	C-24.4 Declaration and Termination of Emergency		
477.	Is the list of designated officials who are to be communicated about declaration and termination of emergency available in the control room / emergency control centre?	The list of designated officials who are to be communicated about declaration and termination of emergency available in the control room / emergency control centre. (Ref.: On-site emergency plan)	---
478.	Are the methods of communication (siren, public address system etc.) for declaration and termination of an emergency known to all the workers?	Siren was available.	---
	C-24.5 Resources-evacuation / Transport		
479.	Are the following resources (equipment, personnel and procedures) required to handle emergency available? 1. Communications, 2. Public announcement systems 3. Monitoring of hazardous releases into the environment, 4. Emergency shelters at the facility, 5. Emergency exits with proper illumination, with uninterrupted power supply, 6. Direction for emergency exit / escape route marked in haulage / Alleyways, 7. Transport for	Following resources (equipment, personnel and procedures) available for handling emergency. 1. Siren, Landline Phone, etc. 2. Public Address System was not evident. 3. Environment and Workplace monitoring records was not evident. 4. It was informed that emergency shelters available. 5. Emergency Exit provided and details available with EHS dept. 6. Assembly Points, Escape route,	226. PA System was not installed. 227. Direction for emergency exit / escape route should be marked in haulage / Alleyway. 228. Workplace monitoring to be carried out.

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	evacuation of plant personnel, 8. Medical care including administration of antidotes, and 9. Security / maintenance of law and order.	emergency exit marked. 7. Transport vehicle available. 8. For information see element C-9, C-22.3 9. Security staff available.	
	C-24.6 Communication Facilities		
480.	Does the emergency control centre have direct communication links with the fire station and the plant control room?	Landline phone, mobile, etc. available for emergency communication.	---
481.	Are there adequate alarm points from which an emergency alarm can be raised?	Siren available at security	---
482.	Is there infrastructure available for ensuring backup electric power supply for communication links where required	DG set available for ensuring backup electric power supply for communication links where required.	---
	C-24.7 Medical Care		
483.	Is the procedure for emergency medical care available?	Procedure for emergency medical care available in On-Site Emergency plan. Mutual aid with nearby hospitals - Jayaben Modi Hospital, Ankleshwar	---
484.	Whether the system has been tested at regular frequency through mock drill / exercises for its adequacy?	Training given	229. Regular mock drill to be carried out.
485.	Does the system of periodic replacement of antidotes and medicines required in emergency exist?	System available for replacement of medicine.	---
	C-24.8 Updation of Emergency Plan		
486.	Is the emergency plan updated based on the feedback from the periodic drills / exercises?	Emergency plan (On-site emergency plan) available and updated as and when required	---
487.	Are the contact details of all concerned officials kept updated in the emergency plan?	The contact details of all concerned officials kept updated in the emergency plan	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	C-24.9 Periodic Drills / Exercises		
488.	Are mock exercises conducted at stipulated intervals?	Training given	230. Regular mock drill to be carried out.
489.	Are the scenarios varied in the mock-exercises to ensure that all possible factors including meteorological conditions, affected plant personnel covered?	Yes Ref.: On-site Emergency plan	---
490.	Whether emergency preparedness Plans have been tested and reviewed at regular frequency through mock drill for its adequacy	Emergency preparedness Plan available	---
	C-24.10 Training of Plant Personnel		
491.	Are the plant personnel trained in handling emergency equipment?	Most of plant personnel trained in handling emergency equipment.	---
	C-24.11 Public Awareness Programme		
492.	Are public awareness programs conducted for the people around the site regarding the actions to be taken in case of off-site emergency?	It was informed that Public Awareness programs conducted for the people around the site regarding the actions to be taken in case of off-site emergency.	---
	C-24.12 Mutual-aid Program		
493.	Are the types of accidents where external organizations would be involved in remedial actions identified? Are their responsibilities defined?	We came to know that all types of accidents where external organizations would be involved in remedial actions were identified. Their responsibilities were also defined in emergency plan.	---
494.	Is the plant responsible for rendering mutual aid assistance to any other external organizations? Does this assistance effect the plant's emergency preparedness?	Plant responsible for rendering mutual aid assistance to any other external organizations. This assistance did not affect the plant's emergency preparedness	---
495.	Whether the communication channels for mutual assistance	The communication channels (Mobile) for mutual assistance	---

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
	identified and known with and between two organizations?	identified and known with and between two organizations	
	C-24.13 Emergency Control Centre		
496.	Is the emergency control centre located beyond the effective distances of identified emergency scenarios?	Emergency control centre located beyond the effective distances of identified emergency scenarios. One Assembly point evident.	---
497.	If the emergency control centre is located within the effect distance, is it suitably protected that it will be available in case of emergency?	The emergency control centre was located within the effect distance, it is suitably protected that it will be available in case of emergency. One Assembly point evident	---
	C-25 Safety Inspection		
	C-25.1 Inspection Programme		
498.	Are checklists available for inspections? For example, availability of checklists like: a. Handling, Storage and Transportation of hazardous chemicals. b. Electrical hazards; c. Fire safety; d. Hand and portable power tools; e. Machine hazards; f. Lifting equipment; g. Ladders and scaffolding; h. Environmental Monitoring; i. Civil structure; j. Housekeeping; k. Emergency equipment; and l. Gas cylinder and other pressure vessels used /available in the organization.	We came to know that most of said Checklist was available.	231. Ensure that all said checklist available and maintained.
	C-25.2 Safety Related Deficiency (SRD) Report		

Sr. No.	Checkpoints	Current Status / Observation	Recommendations
499.	Are SRDs generated based on the area wise checklists?	SRDs was generated informally during plant round.	232. SRD to be recorded in checklist form.
500.	What is the procedure for resolving the SRDs?	Informal procedure to resolving SRDs.	---
501.	Whether the procedure exists for notification and root cause analysis of non-conformities and action taken on them?	Informal system available	233. SRD to be generated for the improvement.
	C-25.3 Safety Inspection Records		
502.	Are the safety inspection records maintained?	Safety inspection was carried out informally during plant round. Records was not evident.	234. Safety inspection records to be maintained.
	C-25.4 Methodology and Inspection Team		
503.	Is there written procedure for safety inspection?	Informal procedure for safety inspection was available.	---
504.	Whether safety inspection is carried out by a designated team?	Safety department was carried out safety inspection.	---
505.	What is the frequency of safety inspections?	Monthly / quarterly frequency of Safety inspection.	---
506.	Whether an inspection report is generated?	Inspection report was recorded informally.	235. Inspection report record to be maintained formally.
	C-25.5 Compliance of Recommendations		
507.	To whom the recommendations are submitted	Recommendations were submitted to Unit Head & Concern person.	---
508.	Are recommendations of safety inspections complied in time?	We came to know that recommendations of safety inspections complied in time.	---
509.	Is compliance of recommendations sent to top management?	Compliance of recommendation sent to top management.	---
510.	Is compliance of recommendations reviewed by safety committee?	Compliance of recommendations reviewed in the meeting	---
511.	Does top management follows-up the compliance?	Top management follows-up the compliance	---
512.	Is compliance of recommendations sent to top management?	Compliance of recommendation sent to top management.	---